Labour-Market Adjustment and Fiscal Incidence of Trade Exposure in the UK*

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Abstract

How does the UK tax–benefit system distribute the fiscal consequences of a trade-related earnings shock across alternative labour-market adjustments? Using the 2024–25 Family Resources Survey and PolicyEngine UK, we hold sector-specific earnings losses calibrated from the 2025 US tariff schedule fixed while varying their incidence across incumbent wage reductions, displacement, transition-equivalent re-employment penalties, and mixed paths. We quantify disposable income, poverty, inequality, and the Exchequer balance by tax and benefit channel. In the organizing full-schedule transition path, the annual earnings loss is £881 million; the tax–benefit system absorbs 41.4 per cent, implying an Exchequer cost of £461 million. The polar wage-cut and displacement specifications provide bounds; sensitivity analysis varies adjustment shares, duration-equivalent penalties, worker selection, and benefit take-up. Fiscal incidence depends substantially on the labour-market margin through which an aggregate earnings loss is realised.

Keywords: tariffs, trade shocks, microsimulation, poverty, inequality, public finances

JEL codes: F13, F16, H23, I38, D31

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1 Introduction

Who bears a trade shock when the same aggregate labour-income loss can be distributed through different labour-market margins? This paper studies that question for the United Kingdom. We hold a trade-exposure-calibrated loss in employee earnings fixed and compare incumbent wage reductions, temporary non-employment, re-employment with an earnings penalty, and mixtures of these paths. The estimand is conditional: it is the statutory tax–benefit and distributional response to a specified earnings-loss path, not a causal estimate of the effect of a particular tariff episode.

This normalization is useful because the UK tax–benefit system is nonlinear. For an equal aggregate loss, a broad reduction in earnings can leave most households in work and within the tax schedule, whereas a concentrated loss can move households across benefit eligibility thresholds. The fiscal incidence therefore depends on who loses earnings, whether the loss is temporary, and whether employment resumes. The comparison is also a warning against interpreting a single wage-cut or displacement scenario as a forecast of realised trade adjustment.

Our design follows the trade-adjustment and automatic-stabiliser literatures. The empirical literature documents persistent non-employment, lower earnings after re-employment, sector switching, and household responses to import competition. Automatic-stabiliser studies then evaluate how taxes and transfers absorb income shocks conditional on these labour-market outcomes. We combine these insights in a transparent scenario surface: pure wage-cut and pure displacement cases are reported as bounds, while mixed paths vary the shares assigned to temporary non-employment, survivor earnings reductions, and post-displacement re-employment penalties. The mixed shares are disciplined by external evidence but are not estimated from the available data.

The 2025 US tariff schedule supplies sector weights and plausible shock magnitudes. We combine tariff rates with US export exposure by SIC division and an export-demand parameter to construct a calibration. This first stage is deliberately not interpreted as identified tariff transmission: a complete trade model would also include quantities, prices, imported inputs, domestic demand, productivity, profits, and firm entry and exit. The HMRC destination panel and longitudinal-LFS exercises are therefore validation and calibration diagnostics, not causal evidence for the main fiscal-incidence estimand.

The second stage uses the 2024–25 Family Resources Survey and PolicyEngine UK to compute taxes, National Insurance, benefits, household disposable income, poverty, inequality, and Exchequer effects. We report a common aggregate loss under alternative adjustment paths, decompose the result by tax and benefit channel, and show sensitivity to duration, worker selection, benefit take-up, and re-employment. Pension contributions are separated from current-income insurance because lower saving can mechanically raise measured disposable income while reducing future resources.

At the full-schedule transition calibration, the common annual earnings loss is £881 million, cushioning is 41.4 per cent, and the Exchequer cost is £461 million. The polar wage-cut and displacement cases remain useful bounds. These are statutory comparisons conditional on the common-loss normalization, not estimates of the tariff's household effect. The paper's central object is the response surface around the bounds, making explicit how fiscal consequences change with temporary non-employment, re-employment and incumbent earnings adjustment.

The paper makes three contributions. First, it provides a reproducible UK microsimulation framework for mapping trade-exposure-calibrated earnings losses into fiscal and distributional outcomes. Second, it shows why the incidence margin matters even when aggregate lost earnings are identical. Third, it quantifies which statutory channels drive the differences and which assumptions—timing, take-up, pensions, and worker selection—are most important for policy conclusions. The remainder of the paper describes the evidence and scenario design, presents the polar bounds and mixed paths, reports policy counterfactuals, and discusses limitations and reproducibility.

2 Literature

This paper sits at the intersection of four literatures: the quantitative and reduced-form work on the incidence of trade shocks; the evidence on how workers actually adjust to them; the institutional analyses of the 2025 US tariffs on the UK; and the microsimulation literature on tax–benefit cushioning of earnings shocks. Existing macro–micro studies impose labour-market shocks on tax–benefit models, while tariff studies map price and income exposure into heterogeneous household welfare. The World Bank's Household Impacts of Tariffs framework combines household consumption and income portfolios (Artuc et al., 2021); the OECD links its METRO trade model to household-budget surveys through an explicit GTAP–COICOP concordance (Luu et al., 2020). The narrower contribution here is a UK application that maps the 2025 partner-country export-demand exposure into alternative labour-income stress scenarios and then through the statutory tax–benefit system. It is a first-round scenario exercise, not a causal estimate of the tariffs' total household incidence.

2.1 Trade shocks and their incidence

Two recent papers anchor the structural end and frame the adjustment-margin question as its poles. Ignatenko et al. (2025) build a quantitative general-equilibrium model—nesting Armington, Eaton–Kortum, Krugman and Melitz structures with exact hat algebra over 194 countries—in which the 2025 tariffs enter as a *primitive* on trade shares. Wages are flexible and labour supply elastic, so employment is an equilibrium *outcome*; strikingly, its sign flips with revenue recycling (a tariff-financed income-tax cut raises US employment by 0.32 per cent, a lump-sum rebate lowers it by 0.41 per cent), and the optimal US tariff is a uniform 19 per cent independent of bilateral deficits. The UK is in their sample, and their replication data provide the UK-specific welfare and employment cells we use as aggregate anchors. Atkin et al. (2025), by contrast, apply a Baqaee–Farhi wedge framework to Chilean administrative data with factor supplies *fixed*: there is no employment margin at all, all adjustment runs through flexible wages and prices, and household incidence arrives via consumption baskets, factor ownership, profits and transfers—their "statutory versus economic incidence" framing is the one we adopt. One paper derives employment from the tariff primitive; the other assumes it fixed and lets wages absorb everything. A third structural anchor sits between the poles: Rodríguez-Clare et al. (2025) embed the 2025 trade war in a dynamic trade-and-reallocation model with *downward nominal wage rigidity*, in which higher tariffs raise US manufacturing employment but lower total employment as falling real wages depress participation, with US real income down about 1 per cent by 2028—rigid wages turn the same primitive into an employment shock that flexible-wage models turn into a wage shock, which is precisely the disagreement our adjustment-margin axis makes an explicit scenario dimension. On the partner-country side, Michelena et al. (2025) run the 2025 escalation through a multiregional input–output framework and find employment and export losses concentrated in the *targeted* countries—the same statutory-versus-economic distinction at the country level, but at the level of sectoral aggregates rather than households.

Complementing the model-based work, den Besten and Känzig (2025) construct a narrative series of US tariff shocks over 1840–2024 and find tariff increases robustly contractionary—imports fall sharply, exports decline with a lag, and output drops persistently—historical discipline for treating the shock as a negative export-demand shock rather than a benign relative-price change.

The reduced-form lineage begins with Autor et al. (2013): import competition caused large, persistent employment and wage losses in exposed US commuting zones, with effects persisting to 2019 and no offsetting migration (Autor et al., 2021). Caliendo et al. (2019) provide the structural counterpart—a dynamic spatial model with costly mobility in which the China shock is a productivity/trade-cost primitive, generating aggregate gains alongside concentrated regional losses—and Adão et al. (2023) supply the theory-consistent mapping from regional shift-share estimates to aggregate counterfactuals that cautions against reading regional results as national ones. Kim and Vogel (2021) translate reduced-form employment and wage responses into money-metric worker welfare, finding regressive incidence with displaced workers' losses exceeding wage losses; Galle et al. (2023) show in a Roy-model setting that aggregate gains coexist with group-specific losses. Our contribution relative to this structural incidence work is to replace stylised worker groups and stylised transfers with real households and the actual UK tax–benefit system.

The 2018–19 US tariff episode supplies the sharpest direct evidence on who pays a tariff. Pass-through into US import prices was essentially complete, placing the incidence on the importing country's firms and consumers (Amiti et al., 2019; Fajgelbaum et al., 2020), with retail prices adjusting more slowly than border prices (Cavallo et al., 2021)—a pattern the 2025 wave repeats: matching daily retailer microdata to product-level tariff rates, Cavallo et al. (2025) estimate retail pass-through of about 20 per cent by September 2025, contributing roughly 0.7 percentage points to the US CPI; and Flaaen and Pierce (2019) show that US manufacturing employment fell on net, as input-cost increases and foreign retaliation outweighed import protection. The retaliation side of that episode is the closest analogue to our channel—a partner's tariff arriving as an export-demand shock: Waugh (2019) finds Chinese retaliatory tariffs depressed consumption in exposed US counties, and Blanchard et al. (2019) trace the electoral fallout of retaliatory tariffs across US counties. The broader distributional literature—surveyed in Fajgelbaum and Khandelwal (2016) and Fajgelbaum and Khandelwal (2022), with Borusyak and Jaravel (2021) decomposing expenditure against earnings channels—establishes that trade shocks redistribute along both consumption and income margins. The narrower contribution here is to apply a UK statutory tax–benefit microsimulation to alternative labour-income stress scenarios; it does not claim that the broader trade-and-household intersection is empty. Two propagation channels our direct-channel design deliberately excludes have well-measured analogues in this literature. Firm-level shocks travel along supply chains—Barrot and Sauvagnat (2016) show idiosyncratic disruptions imposing substantial output losses on customers of affected suppliers, and Acemoglu et al. (2016) find network effects on aggregate outcomes several times larger than the direct effect—and tradable-sector job losses can affect local services (Moretti, 2010; Gathmann et al., 2020). These estimates are not mechanically composable with our accounting sensitivity; Section 6 carries the caveat. Open-economy macro adds an offsetting margin: exchange rates absorb part of a tariff, but imperfectly and state-dependently—at most a fifth of the 2018–19 dollar appreciation is attributable to the tariffs (Jeanne and Son, 2024), the dollar appreciates on unilateral tariffs but depreciates under retaliation (Corsetti et al., 2025), and tariff *uncertainty* itself depreciated the dollar in 2025 through a risk-premium channel (Kalemli-Özcan et al., 2026); model-based analyses of retaliation similarly find non-trivial consumption losses for passive trade partners (Auray et al., 2020). We hold the exchange rate fixed; no scenario parameter is claimed to absorb or identify that response. Early assessments of the 2025 wave itself (Fajgelbaum and Khandelwal, 2026) again find the incidence landing on the tariff-imposing country's consumers—complementary to our question, which is the incidence on the tariffed *partner's* workers.

2.2 Adjustment margins: how workers absorb trade shocks

The worker-level evidence says the margin is mostly earnings, not permanent unemployment. Autor et al. (2014) find US workers exposed to import competition lose earnings amid churn across employers and sectors rather than through indefinite joblessness. Traiberman (2019) shows for Denmark that occupation-specific human capital produces reallocation with persistent wage losses; because our shock is an *export-demand* shock rather than import competition, the most direct microeconometric precedents are Hummels et al. (2014), who instrument Danish firm-level export demand and trace it into worker wages by task, and Garin and Silvério (2024), who show that Portuguese exporters pass idiosyncratic export-demand shocks into the wages of incumbent workers rather than shedding them—evidence for including a wage-cut dimension in our scenario grid. The rent-sharing literature disciplines the plausible scale of incumbent wage responses: Card et al. (2018) survey elasticities of wages to firm-specific rents of 0.05–0.15, and Kline et al. (2019) estimate that workers capture roughly 30 per cent of patent-induced surplus, with gains concentrated among incumbent men. Against that evidence, our pure wage-cut and pure displacement margins are polar stress scenarios. The 15/85 mixed point of Section 3.4 is literature-disciplined but is not an empirical conversion of those elasticities into adjustment shares. Dauth et al. (2021) provide the cleanest evidence for the wage-margin family, with German workers reallocating into services and other plants at lower wages. Dix-Carneiro (2014) estimates that reallocation takes years, with adjustment costs of 1.4–2.7 times annual wages and older workers hit hardest—calibrating our transition horizons and age gradients—while Dix-Carneiro and Kovak (2017) show regional effects that *grow* over two decades on a joint wage/employment margin.

The UK's own history points to a third margin absent from the US and German studies: benefit-financed economic inactivity. Beatty et al. (2007) document that coalfield job losses became male inactivity on incapacity benefits, not measured unemployment, with incomplete recovery after twenty years; Beatty and Fothergill (2017) show that 1980s–90s industrial job destruction *still* inflates the deficit through higher benefit claims and lower tax receipts—the direct UK precedent for our fiscal-incidence question, retrospective and area-level where ours is prospective, rules-based and household-level. The margin is not merely historical: Roberts and Taylor (2022) show in 2009–18 longitudinal data that disability benefit claims still respond to local labour-market slack over and above health status, Beatty and Fothergill (2023) document the persistence of hidden unemployment among incapacity claimants across large parts of Britain, and the IFS records 4.2 million working-age people on health-related benefits by 2024, up from 3.2 million in 2019 (IFS, 2024)—the inactivity margin is live at exactly the moment the tariff shock arrives. Aragón et al. (2018) add gender-differentiated margins from mine closures, and Rud et al. (2022) decompose displacement costs into wage cuts on re-employment versus non-employment spells—directly calibrating the split between our displacement and wage-cut families. For UK-specific elasticities, De Lyon and Pessoa (2021) is the key source: UK workers in Chinese-import-competing industries lost employment years and earnings without equally gainful re-employment, with larger female losses through employment years (see also Dorn and Levell, 2024). Closest to our object, Irastorza-Fadrique et al. (2025) study UK *household* responses to trade shocks—added-worker effects, later retirement, self-employment—but model behaviour, not tax–benefit mechanics; we treat their margins as complementary to our rules-based incidence.

2.3 The 2025 tariffs and the UK

The institutional literature on the 2025 shock is rich but stops above the household. NIESR's NiGEM scenarios and the OBR's March 2025 tariff analysis provide the macro anchors—the OBR puts the permanent UK GDP loss from a 20-point US tariff at one-third to three-quarters of a per cent (NIESR, 2024; OBR, 2025). The Bank of England estimates the effective US tariff on the UK at roughly 9 per cent (against about 18 per cent globally) and finds trade diversion has mildly *lowered* UK import prices (Bank of England, 2025; Dhingra, 2025)—which is what licenses our isolating the export-demand/earnings channel while holding the price and monetary channels fixed; the Bank's own staff work on the exchange-rate response to tariffs and retaliations (Corsetti et al., 2025) supplies the FX side of that held-fixed channel. The multilateral institutions frame the aggregates: the IMF's October 2025 outlook has global growth slowing to 3.2 per cent in 2025 amid the tariff shock (IMF, 2025), and the WTO's October 2025 update puts merchandise trade volume growth at 2.4 per cent in 2025 but only 0.5 per cent in 2026 as the tariffs bind (WTO, 2025). Firm surveys find most UK firms expecting small negative effects, with the US taking about 22 per cent of UK exports (CEP, 2025). The regional and sectoral geography is sharp: Coventry sends 22 per cent of its exports to the US (Centre for Cities, 2025); auto tariffs are estimated to cost £10,000 million of GDP over five years with the West Midlands bearing 62 per cent of it (City-REDI, 2025); SMMT data show US vehicle exports collapsing from 7,077 units in April to 4,223 in May 2025 before the EPD's implementation (SMMT, 2025). UK Steel notes that some 80 per cent of UK steel exports go to Europe—the basis of our steel caveat—and the December 2025 pharmaceutical deal (0 per cent tariffs for three years, paid for through VPAG rebate cuts from 22.5 to 14.5 per cent and NHS pricing concessions) makes pharma a fiscal/pricing shock rather than an employment shock (ABPI, 2025). The Resolution Foundation's household-facing analysis is descriptive rather than microsimulated (Resolution Foundation, 2025a); the IFS has no dedicated distributional publication on the 2025 tariffs. The closest work leaves room for a narrower UK statutory tax–benefit stress-test contribution.

For UK trade-shock precedents more broadly, Dhingra et al. (2017) map Brexit trade costs to income losses, and Fetzer and Wang (2024) provide the best district-level trade-shock incidence estimates (synthetic-control output losses of 5–10 points concentrated in manufacturing districts); Fetzer (2019) and Bloom et al. (2019) supply the political-economy and uncertainty channels. On household prices, Breinlich et al. (2022) use product-level import shares to identify the pass-through of the referendum depreciation, estimating a 2.9 per cent consumer-price increase and a similar real-wage loss. Levell et al. (2017) show why incidence can differ across UK households: food has much greater import content than consumption overall and household budget shares differ. These studies benchmark a consumption-price channel that the present cash-income exercise deliberately holds fixed.

The institutional modelling benchmark is also relevant to what this exercise does and does not claim. The UK Trade Modelling Review recommends a relatively simple, empirically grounded CGE core, complemented by granular partial-equilibrium modules, regional and dynamic extensions, parameter ranges and ex-post validation (Department for International Trade, 2021). In that architecture the tariff is a policy primitive: relative prices and trade quantities adjust first, production and sectoral value added respond next, and factor demand, wages and employment emerge jointly with reallocation across activities. Firm heterogeneity adds selection and productivity responses, while input–output structure transmits intermediate-cost and demand shocks. The OECD METRO model embodies this sequence in a multi-sector, multi-region production system before mapping induced prices into household expenditure and compensating variation (Luu et al., 2020). The World Bank instead offers transparent first-order household accounting of consumer and producer-income exposure, while explicitly excluding substitution, variety and productivity responses that can dominate beyond the short run (Artuc et al., 2021). Relative to those economy-wide or price-side tools, this paper supplies only the terminal tax–benefit module: it conditions on an imposed labour-income distribution rather than deriving it from production, productivity or labour demand.

2.4 Tax–benefit cushioning of earnings shocks

The methodological ancestors are Dolls et al. (2012), who use EUROMOD and TAXSIM to show EU tax–benefit systems absorb on average about 38 per cent of a proportional income shock and 47 per cent of an unemployment shock (against roughly 32 and 34 per cent respectively in the US). The EU averages are not the right comparator for this paper; their *UK* cells are, and they are close to ours: the UK income-shock stabilisation coefficient is 0.352 (income tax 0.267, social insurance contributions 0.054, benefits 0.031) and the UK unemployment-shock coefficient 0.415 (tax 0.191, SIC 0.061, out-of-work benefits 0.163). Our 43.9 per cent wage-cut rate and 34.1 ± 5.2 per cent displacement rate are of a similar order, which is a useful benchmark rather than an external validation test. Section 4.2 instead focuses on the *composition* of the stabilisation total and the exposed-worker population. The interpretation of the tax term as an effective *marginal* rate is theirs too—and older still: Auerbach and Feenberg (2000) originate the framing of the income tax as an automatic stabiliser operating at marginal rates, and McKay and Reis (2016) formalise how progressivity and the extensive/intensive structure of shocks determine stabilisation in general equilibrium. The asymmetry between marginal and average relief is therefore not our discovery; our application shows how that asymmetry operates for the selected exposed-sector population. A second ancestor is Sutherland and Figari (2013) on EUROMOD's stress-test use—but both apply stylised macro shocks, not trade-policy-derived ones. Our direct methodological sibling is Brewer and Tasseva (2021), who pass the Covid-19 labour-market shock through UKMOD to show that the UK tax–benefit system plus the emergency support schemes held the rise in income inequality far below the rise in earnings inequality: the same architecture—an externally derived shock realised on survey households and run through the full UK benefit rules—applied to a pandemic where ours applies it to a trade-policy shock, without the emergency schemes and with the adjustment margin as the central axis. The wider macro-micro linkage literature supplies the lineage for the shock-derivation step itself: Bourguignon et al. (2008) codify the techniques for passing macro shocks to household microdata, Peichl (2016) surveys the linkage of microsimulation to structural macro models, and Navicke et al. (2013) show how EUROMOD nowcasts impose estimated labour-market transitions on survey microdata—the same device as our displacement draws, applied there to observed rather than counterfactual shocks. On the US side, exposed commuting zones saw rising disability, retirement and welfare payments that offset only a small fraction of earnings losses (Autor et al., 2013); Hyman et al. (2024) show wage insurance raises re-employment and earnings, a candidate counterfactual reform for this framework. The nearest tariff-microsimulation links are on the US side: the Tax Policy Center's tariff models allocate tariff burdens to tax units through consumption baskets in a full microsimulation model (Tax Policy Center, 2025), the Budget Lab at Yale prices the 2025 schedule's fiscal and distributional effects by income decile (Budget Lab, 2025), and Acosta and Cox (2024) document that the US tariff code is built regressively, with higher rates on lower-end varieties of the same goods. Two institutional facts motivate the cushioning asymmetry examined here. UC's capital rules—thresholds frozen in cash terms since 2006—reduced or extinguished entitlement for around two million income-eligible families, with roughly 1.1 million standing to gain from uprating alone (Resolution Foundation, 2025b); and take-up of income-related benefits is well below 100 per cent (DWP, 2024), so statutory replacement rates can overstate what displaced workers receive. Take-up gates entitlement inside the model and is explored only in a five-assignment sensitivity grid (Section 3.6), while the capital rules are a real-world gate the plain FRS data cannot exercise (Section 6). The distinction between consumer-price incidence and a partner's export-demand/earnings incidence defines the paper's narrower contribution: a UK statutory fiscal stress test organised around alternative adjustment margins. The results do not overturn the automatic-stabiliser literature; they show how the tax and benefit components differ across imposed margins for this exposed-sector population.

3 Methodology

3.1 Design: primitives first

Throughout, "derived" means generated by the stated scenario mapping, not causally estimated. The central demand parameter is a transparent unit stress, $\varepsilon = 1.0$, rather than an estimate. The grid includes $\varepsilon = 0.4$ as an OBR-style lower anchor, 2.0 as the former April-outturn high case, and 3.0 as a severe sensitivity. The April movement is not used to identify the centre because it also reflects anticipation, front-running, prices, exchange rates, composition and other demand conditions. Likewise, $\pi = 1.0$ is a deliberately strong full wage-bill-incidence assumption. The parameter grid is scenario sensitivity, not a statistical confidence interval.

A reader is entitled to ask, before any number is reported, which parts of this design are read off data and which are assumed. The answer, stated in full:

Object	Status
Tariff schedule τ_j	**Data** (policy documents: the April 2025 schedule, the May 2025 EPD, the December 2025 pharmaceutical agreement).
US-export intensity x_j	**Data** (HMRC RTS/OTS customs exports over ONS ABS division turnover; Section 7).
Export fall	**Data** in the measured family (realised HMRC outturn, May 2025–February 2026); **assumed** in the tariff families through declared low, central, high and severe demand scenarios.
Pass-through π	**Assumed** ($\pi = 1.0$), varied on a grid (Section 7).
Adjustment margin	**Assumed**, and deliberately varied. This is the paper's free dimension.
Household incidence	**Computed** from statutory tax and benefit rules in PolicyEngine UK; no behavioural or reduced-form step.

The margin is assumed because the literature does not agree on it, and the disagreement is structural rather than empirical: Ignatenko et al. (2025) derive the employment response from elastic labour supply and flexible wages, while Rodríguez-Clare et al. (2025) derive it from downward nominal wage rigidity, so the same tariff primitive arrives as a wage shock in the first model and a job shock in the second. We do not endogenise the margin. We price the disagreement—holding the aggregate earnings loss fixed and reporting what each pole implies for households and the Exchequer. That is a weaker claim than a structural model makes, and a more auditable one.

The shock enters through primitives, not through an assumed employment effect. The pipeline is:

$$s_j \;=\; \tau_j \, x_j \, \varepsilon \, \pi, \tag{1}$$

where s_j is an imposed employee-wage-bill stress for SIC 2007 division j, τ_j is its tariff rate, x_j is US-export intensity, ε is an export-demand calibration and π is a *wage-bill-incidence assumption*. Calling π pass-through would conflate distinct mechanisms. A structural transmission would separately map the tariff into export prices and quantities, quantities into domestic production and value added, and production into factor demand, productivity, profits, wages, hours and employment. Equation (1) compresses those links into one accounting bridge. It neither estimates a production function nor identifies the causal response of productivity, output or labour demand.

x_j is US goods exports as a share of division turnover, built from HMRC trade data and ONS Annual Business Survey turnover denominators; Section 7 documents the frozen provenance and denominator choice. The central $\varepsilon = 1.0$ is a transparent unit stress, not an elasticity estimate; it amounts to a unit tariff-to-export-demand semi-elasticity, under which each percentage point of tariff removes one per cent of US-bound demand, and $\varepsilon = 0.4$ is the lower anchor, in the spirit of the trade-elasticity assumptions underlying the OBR's March 2025 tariff scenarios (OBR, 2025). The central $\pi = 1$ assigns the full calibrated exposure to the employee wage bill; it is intentionally severe and is not established by the economics literature. The sensitivity grid varies $\varepsilon\pi$, but this is parameter sensitivity, not an uncertainty interval around an identified coefficient. The implied gross earnings loss is therefore a conditional scenario input. OECD METRO and the UK Trade Modelling Review instead place tariff shocks inside multi-sector production systems in which prices, output, value added and factor demand adjust jointly (Luu et al., 2020; Department for International Trade, 2021); the World Bank HIT framework similarly labels its direct household calculations first-order and excludes productivity and substitution responses (Artuc et al., 2021). This paper begins only after those missing production-side mechanisms would have generated a labour-income distribution.

3.2 Public product–destination benchmark

To test whether post-policy US exports move differently from other UK export destinations, I build a monthly HMRC OTS panel from January 2018 to May 2026 for the United States, Canada, Japan and Australia (HM Revenue and Customs, 2026). Detailed HS8 rows are aggregated to HS4 and balanced across products, destinations and months, with absent flows set to zero. HMRC's separate two-digit aggregate rows are excluded. For product p and month t, the outcome is

$$g_{pt} = \log(1 + V_{p,\mathrm{US},t}) - \frac{1}{3} \sum_{d \in \{\mathrm{CA},\mathrm{JP},\mathrm{AU}\}} \log(1 + V_{pdt}).$$

I regress this destination gap on a post-May-2025 indicator, HS4 product fixed effects, month-of-year effects and a linear trend. April 2025 is omitted as an announcement and anticipation month. The primary WLS weights are products' mean 2022–24 US export values, capped at the 99th percentile; standard errors are clustered by HS4. Uncapped value weights, equal product weights, individual destinations, later sample starts and pseudo-policy dates in May 2019–23 are reported as specification checks.

This design controls for product shocks common across destinations but not US-specific demand, exchange rates, shipping, inventories, composition or other coincident policies. The high-tariff comparison groups HS chapters 72, 73 and 87 as steel and automotive products; that broad mapping is a falsification check, not a complete product-level statutory tariff schedule. The panel is therefore not used to estimate ε or π in Equation (1). It tests whether the imposed bridge is consistent with destination-differential outturn patterns.

3.3 The tariff schedule and the EPD counterfactual

Two tariff scenarios bracket the policy. *Full tariffs*: the April 2025 schedule—10 per cent baseline on UK goods, 25 per cent on motor vehicles (SIC 29) and steel (SIC 24). *EPD*: the deal-mitigated schedule—autos at the 10 per cent in-quota rate on up to 100,000 vehicles; steel with conditional, partial relief; pharmaceuticals (SIC 21) exempt under the December 2025 agreement. The automotive scenario applies the 10 per cent rate to the modelled 2024 exposure because the quota is approximately equal to that year's US-bound volume; it does not model quota allocation, above-quota growth or an endogenous volume response around the ceiling. The difference between the two scenarios, run through identical downstream machinery, prices the EPD for households and the Exchequer. Two caveats travel with the schedule. Steel: about 80 per cent of UK steel exports go to the EU, so the US tariff is a minority driver of the sector's exposure and the steel cell should be read alongside EU/UK safeguards. Pharmaceuticals: the exemption was purchased through VPAG rebate cuts and NHS pricing concessions, so pharma's incidence is fiscal and price-side, not an employment shock; we model its employment shock as zero under the EPD and discuss the fiscal channel separately (Section 5).

3.4 Adjustment-margin families

The derived s_j is realised on Family Resources Survey 2024–25 employees in exposed divisions. The submission design treats the mixed adjustment surface as the conceptual centre: statutory cushioning is evaluated for a common expected sector wage-bill loss while the share delivered through temporary non-employment, survivor earnings reductions, and re-employment is varied. The pure displacement and proportional wage-cut cases are transparent bounds on that surface. Inactivity, literal sectoral reallocation and the EPD schedule are complementary sensitivities, not alternative causal estimates. Wage cuts remove $\sum_j s_j B_j$ deterministically; displacement removes that amount in expectation under the risk model and is numerically integrated using the balanced design below.

Displacement (ADH short run). Within each exposed division every employee has baseline risk s_j, irrespective of the record's survey weight. The primary estimator generates repeated random-start systematic candidates, ordered across age and within-industry earnings ranks, and retains the candidate that best matches the expected industry wage-bill and weighted- head-count losses while secondarily balancing broad age and earnings margins. This is balanced numerical integration, not a survey estimator: conditioning on aggregate balance intentionally changes exact record-level marginal inclusion probabilities, and because the retained candidate minimises a score dominated by the total wage-bill target, per-division losses are matched softly rather than exactly and best-of-candidates selection carries no formal unbiasedness guarantee.

The claim that the balanced design removes the declared amount in expectation is therefore established empirically, by its agreement with the unbiased comparator, not by construction. Selected workers move to unemployment with earnings, hours, pension contributions and statutory pay zeroed. An independent Bernoulli estimator, which preserves the declared first-order probabilities exactly but has much larger allocation dispersion, is reported as the unbiased assignment-rule comparator; at the unit scale the two agree on cushioning to half a percentage point (Section 4.2).

Wage cuts with reallocation (Dauth/Traiberman long run). No job loss; every employee in division j takes a proportional cut of s_j, so the weighted aggregate earnings removed equals $\sum_j s_j B_j$ by construction—exact earnings-equivalence with the displacement family's expected removal. Employee pension and salary-sacrifice contributions are scaled proportionally with the cut, mirroring their zeroing under displacement; this lowers taxable pay (and hence tax relief) but also reduces the household's pension outflow, so under the HBAI disposable-income concept part of the earnings loss is absorbed by reduced retirement saving rather than by current income.

Inactivity (Beatty–Fothergill UK history). The displacement draw, but displaced workers aged 50 and over exit to economic inactivity, the benefit-financed route of UK deindustrialisation, rather than unemployment. The route is benefit-financed in the model, not just relabelled: every inactive worker is flagged as having limited capability for work-related activity, so their household's Universal Credit includes the LCWRA health element (£217.26 a month in 2026—the reduced rate applying to new claims from April 2026 under the Universal Credit Act 2025; existing claimants retain the higher pre-reform rate, and all modelled inactivity claims are new claims). In policyengine-uk 2.89.2 the survey-imputed `uc_LCWRA_element` is stored, so the post-shock benefit-unit value is set to the maximum of the stored amount and one annual statutory element; an existing recipient is never paid a second element, unaffected units are unchanged, and the run hard-errors if the element fails to reach a newly flagged person's benefit unit. This mirrors the Beatty et al. (2007) route by which industrial job loss became incapacity-benefit inactivity. The strong assumption should be stated plainly: *all* displaced workers aged 50 and over are assumed to pass the Work Capability Assessment, so the family is an upper bound on the health-element route; no partial-qualification sensitivity (for example, 50 per cent of the older displaced qualifying) has been simulated, which is a limitation of the present scenario set. Work-search conditionality carries no separate fiscal machinery in the model, so the LCWRA element is the entire modelled wedge between inactivity and unemployment.

Reallocation into services (ADH/Autor et al., 2014). The same displacement draw on the same seed—so the two families are paired worker-for-worker—but the drawn workers are re-employed rather than left out of work. Their `sic_industry_division` is reassigned to one of four services destinations (SIC 47 retail, 86 human health, 49 land transport, 56 food and beverage service) in proportion to those divisions' FRS employee-weight shares (29.5, 45.8, 9.2 and 15.5 per cent), and their annual earnings are cut by a penalty calibrated on the FRS itself: weighted mean annual employee earnings of £48,272 over 2,047 hours in the exposed goods divisions against £34,594 over 1,701 hours in the four destinations, a **28.3 per cent annual-earnings penalty** that embeds the hours fall. Two variants are reported: an hours- and age-controlled **low-penalty** case of 14.0 per cent (the pure hourly-wage gap), and a **three-**

month lag in which workers are out of work for a quarter of the year before the services job starts, scaling annual earnings and hours by a further $1 - 3/12$. ASHE 2024 full-time medians bracket the hours-controlled figure at 10–20 per cent; the FRS all-employee penalty is larger because it also captures the shift into part-time work. Because this family removes only the penalty on the movers rather than the full earnings-equivalent aggregate, its pound figures are comparable with the displacement family (identical worker sets, paired draws) but *not* with the wage-cut family; its cushioning *rate* is comparable with both. Section 7 reports the results and mechanics.

Common-loss transition path. The redesigned transition margin combines a transition-equivalent annual earnings/hours penalty, service-sector re-employment and survivor earnings reductions while preserving the common-loss normalization. Let s be the sector shock, λ the share of the declared loss assigned to transition workers, and $d = 1 - (1 - r)(1 - \ell/12)$ the annual earnings loss of a transition worker, where r is the re-employment penalty and ℓ the lag in months. Transition probability is $p = \lambda s/d$; survivors take a cut $c = (1 - \lambda)s/(1 - p)$. Hence $pd + (1 - p)c = s$ in expectation. The organizing calibration uses $\lambda = 0.70$, $r = 0.283$ and $\ell = 3$. In the annual simulation, the lag is represented by an earnings/hours factor: workers remain employed for PolicyEngine purposes, so no within-year unemployment or JSA/UC spell is generated. This is an accounting normalization for comparing paths, not an estimate of the probability of re-employment or the UK tariff response. The 14 per cent hours/age-controlled penalty and zero-, three- and six-month lags are sensitivity dimensions.

Mixed adjustment and factorial scenario testing. Let $\lambda \in [0, 1]$ denote the share of the sector shock delivered through temporary non-employment rather than an incumbent earnings reduction. A worker with sector shock s is displaced with probability $p = \lambda s$. Conditional on surviving, the proportional wage cut is

$$c = \frac{s - p}{1 - p},$$

so expected earnings loss is exactly $p + (1 - p)c = s$ for every worker. The endpoints reproduce pure wage cuts ($\lambda = 0$) and pure temporary displacement ($\lambda = 1$); because the mixed family draws displacement independently, the $\lambda = 1$ endpoint reproduces the Bernoulli comparator rather than the balanced primary estimator. A separate reallocation family moves displaced workers into observed service destinations, with either an immediate earnings penalty or a three-month non-employment lag. It is paired worker-for-worker with displacement but does not preserve the common aggregate loss unless its penalty is explicitly re-scaled, so its pound totals are reported as a transition sensitivity rather than combined with the mixed grid. The exploratory grid crosses $\lambda \in \{0, .25, .5, .75, 1\}$ with export-demand calibrations $\varepsilon \in \{0.4, 1, 2, 3\}$ on five common assignments per cell. This follows the companion UK AI study's factorial principle: vary shock magnitude separately from adjustment composition. It is scenario testing, not a probability distribution over future outcomes.

The wage-cut margin as an upper bound, and a literature-disciplined mixed sensitivity. The pure wage-cut family should be read against the rent-sharing evidence, because it embeds a survivor-wage response far above anything estimated. Delivering the entire sector wage-bill loss through proportional cuts to incumbent workers' wages imposes a unit wage response to the sector exposure index. Garin and Silvério (2024) estimate an incumbent-wage elasticity of roughly 0.13–0.14 to firm-specific output shocks in Portuguese matched data; the rent-sharing survey of Card et al. (2018) places typical elasticities at 0.05–0.15; and Hummels et al. (2014) find small within-job wage responses in Danish data alongside losses through displacement. The wage-cut family ($\lambda = 0$) is therefore an *upper-bound stress scenario for survivor-wage incidence*, retained because it prices the flexible-wage pole of the structural disagreement.

The scenario set also reports a transparent 15/85 mixed sensitivity: 15 per cent of the imposed wage-bill loss is delivered through survivor wage cuts and 85 per cent through displacement ($\lambda = 0.85$). The value 0.15 is disciplined by the upper end of the cited wage-response estimates, but it is not an empirical calibration of the loss decomposition. A wage elasticity to firm output or rents is not algebraically the share of an aggregate wage-bill loss borne by survivor wages. Identifying that share requires linked employer–worker estimates of wages, hours, employment and re-employment. The mixed result is therefore a literature-disciplined scenario point, not the uniquely "most plausible" margin.

Two scope limits on the family set should be stated here rather than in a footnote. The wage-cut family models the *earnings consequence* of reallocation without moving anyone between sectors: workers keep their observed industry and hours, and the proportional cut stands in for the wage penalty that a completed reallocation would carry; the reallocation family does move them, but who reallocates is assumed (it is the displacement draw) rather than estimated. And no household-side behavioural margin is active—added-worker responses, earlier or later retirement, and shifts into self-employment (Irastorza-Fadrique et al., 2025) are all suppressed, as are any labour-supply responses to the tax and benefit changes themselves. All four families are therefore first-round, rules-based incidence.

A third caveat concerns selection within divisions. Export-exposed workers are unlikely to be representative of their division: exporters pay a wage premium and pass export-demand shocks into incumbent wages (Hummels et al., 2014; Garin and Silvério, 2024). Cushioning is nonlinear in earnings, so uniform risk may misstate its tax/benefit composition. The longitudinal-LFS cell, earnings-band and QRF sensitivities vary observed risk shape while recalibrating each division to its wage-bill target. They quantify selection- model sensitivity but cannot identify exporter status or tariff-specific worker risk.

The FRS support for the transition is thin. Of 34,966 person records, 12,800 have employee earnings and 1,018 are in mapped exposed divisions, but the reference probabilities imply only 9.7 selected records. The largest exposed record has a person weight above 19 thousand. Balanced integration constrains the realised aggregate and observed composition; it does not create survey information or establish population representativeness. I therefore report the affected record count, Kish-style effective number of gross-loss contributions, largest-record loss share, and balanced-versus-Bernoulli mean and SD comparisons. Subgroup rankings remain exploratory.

Duration-equivalent annual stresses. The annual PolicyEngine model cannot represent a worker employed for part of a year and unemployed for the remainder with internally consistent monthly UC assessment, work status and earnings. I therefore do not simulate three- or six-month unemployment spells. Instead I scale s_j by $d/12$, for $d \in \{3, 6, 12\}$, and assign a correspondingly smaller number of coherent full-year displaced states. These rows preserve the expected worker-month earnings loss associated with each duration while changing the extensive-margin stock. They are labelled *duration-equivalent annual stresses*, not spell estimates. Wage-cut rows use the same scale, preserving the common- loss comparison at every duration and at both the OBR-style 0.4 and unit anchors.

The margin is the paper's central axis because the tax–benefit system treats the two poles asymmetrically: an in-work earnings reduction is offset automatically at marginal tax and taper rates, while a job loss moves the worker to the gated, means-tested extensive margin of the benefit system. How much each pole is cushioned, and—the question that turns out to discriminate between them—how much that cushioning depends on anyone doing anything, are empirical questions the microdata answer (Section 4.2).

3.5 Longitudinal LFS imputation benchmark

The uniform within-division assignment in the central displacement family is transparent but suppresses observable worker heterogeneity. I therefore use UK Data Service Study 9490, the Labour Force Survey Five-Quarter Longitudinal Dataset covering July 2024–September 2025, as a donor benchmark. The donor population comprises respondents employed at wave 1. Job exit is observed at wave 5; log wage change is defined only for respondents employed with positive main-job pay at both endpoints. Negative UKDS sentinel values are missing, not zero. Public 2024 Nomis BRES employee totals rescale the manufacturing SIC composition before direct transition targets are computed. (Office for National Statistics, 2025)

The primary imputation maps SIC-division–sex–age cells to employed FRS receivers, shrinking sparse cells towards sector and manufacturing means. After matching, predicted exit risks are rescaled so their FRS-weighted mean equals the directly measured, BRES-composition-adjusted LFS manufacturing exit rate. Mean predicted log wage changes are centred on the corresponding LFS target. An income-tercile estimator supplies a transparent sensitivity specification.

A regularised quantile/random-forest benchmark follows the calibrated imputation architecture in PolicyEngine's `young-worker-nics` project. It estimates model shape on all wave-1 employed adult donors using age, sex and annualised employment income. The binary outcome uses the random-forest event probability; the continuous outcome uses the QRF conditional mean. Forests use 500 trees and a minimum leaf size of 20; wage outcomes are winsorised at the donor 1st and 99th percentiles. Both receiver predictions are then calibrated to exactly the same manufacturing LFS targets as the cell estimator. This is a predictive baseline-transition benchmark, not an estimate of tariff-induced exits or wage changes. It is therefore reported as a robustness diagnostic and does not replace the declared central assignment rule.

In a downstream allocation sensitivity, each receiver score r_i is converted to a within-division displacement probability $p_i = \min(1, k_j r_i)$, where k_j is solved numerically so

$$\frac{\sum_{i \in j} p_i E_i w_i}{\sum_{i \in j} E_i w_i} = s_j.$$

Thus cells, earnings bands and QRF change who is exposed while preserving each division's expected employee-wage-bill loss. Missing scores use the division mean. Common assignments then isolate sensitivity to incidence shape; they do not give the baseline transition model a causal tariff interpretation.

3.6 Universal Credit take-up after the shock

The take-up experiment is applied symmetrically across margins. After each shock, changed benefit units whose all-claim UC award moves from zero at baseline to positive post-shock receive a claiming draw at the scenario take-up rate. This includes wage-cut households that cross into entitlement. Existing claimants, existing eligible non-claimants, unchanged units and post-shock-ineligible units retain their baseline flag, avoiding an unintended population-wide take-up reform. The grid therefore measures sensitivity to a common new-entitlement convention; it does not estimate actual claiming behaviour.

One modelling choice deserves its own subsection, because an earlier version of this paper got it wrong and the correction changes a headline number. PolicyEngine UK carries benefit take-up as a benefit-unit-level *stored input*, `would_claim_uc`, drawn once at dataset build time. Two properties of that draw matter. First, its target is a flat population-wide parameter—0.55 as a share of *all* benefit units, with reported UC recipients anchored to TRUE and the remainder filled to hit the target. It is not a take-up rate among the entitled, and it is therefore not comparable to the roughly 80 per cent take-up-among-entitled that the Resolution Foundation and the IFS assume; in the dataset used here the realised weighted flag rate is 0.547 over all benefit units and 0.541 among UC-eligible ones. Second, the flag is drawn on *pre-shock* circumstances. Nothing in a naive shock pipeline re-draws it, so a worker displaced by the shock carries into unemployment the claiming decision attributed to them while they were employed and, in the great majority of cases, not entitled to anything—measured take-up among the displaced post-shock was 46.9 per cent, below even the population flag rate. The baseline flag is thus not informative about the post-shock claiming behaviour of a newly entitled family: it encodes a decision taken in a state of the world the shock has destroyed.

The pipeline therefore re-draws `would_claim_uc` after each shock only for benefit units whose circumstances changed and whose all-claim UC award moves from zero at baseline to positive post-shock, at a scenario parameter `uc_takeup` with default 0.80. The re-draw uses an independent random stream, leaving worker assignment unchanged and preserving paired comparisons. This rule applies to displacement, inactivity, reallocation and wage cuts alike; existing eligible, post-shock-ineligible and unchanged units retain their baseline flag. Take-up sensitivity is therefore evaluated symmetrically for newly entitled units, although the parameter remains assumed rather than observed.

3.7 Microsimulation and outcomes

The Monte Carlo draws describe sensitivity to the artificial assignment of displacement across weighted FRS records. Their standard deviations are assignment dispersion, not sampling standard errors, confidence intervals, or uncertainty about the tariff response, survey design or model parameters. Numerical Monte Carlo error of a reported mean is the assignment SD divided by the square root of the number of draws. Parameter scenarios vary the demand calibration, wage-bill incidence, take-up and adjustment composition, but are not assigned probabilities and do not form confidence intervals. Survey sampling variance is approximated for the primary contrast by a household bootstrap of the FRS sample (Section 4.2); the research file carries no PSU or stratum identifiers, so the bootstrap treats households as independent and cannot reflect geographic clustering. Uncertainty in the trade-to-earnings bridge is not estimated. Reported comparisons other than the bootstrapped primary contrast are descriptive scenario contrasts. Distributional cash-income changes use annual household disposable-income change divided by household size before person-weighted aggregation; aggregate disposable-income totals remain household-weighted totals.

The realisation step—an imposed status and earnings transition on survey households, then run through the full benefit rules—follows Brewer and Tasseva (2021), who apply it to the Covid-19 labour-market shock in UKMOD, and the EUROMOD nowcasting tradition of imposed labour-market transitions (Navicke et al., 2013). PolicyEngine UK computes baseline and shocked household disposable income (HBAI cash concept throughout), relative before-housing-costs poverty measured against a poverty line frozen at 60 per cent of the baseline equivalised median (so the reported change reflects households crossing a fixed threshold rather than mechanical movement of the line itself), absolute before-housing-costs poverty against a fixed baseline line, after-housing-costs poverty, the Gini coefficient, distributional breakdowns and the Exchequer effect. The *Exchequer effect* is defined as the change in the aggregate simulated government balance—all modelled tax and contribution revenue (including employer National Insurance contributions) minus all modelled benefit spending—between the baseline and shocked simulations. It is therefore broader than the employee-earnings loss ΔE that enters the cushioning rate; Section 7 reconciles the two. On timing: trade exposure is built from 2024 HMRC customs data, the household input is FRS 2024–25, and the simulation year is 2026, with PolicyEngine UK's standard uprating applied to survey monetary variables between the survey year and the simulation year. Primary submission scenarios use 50 balanced assignments; legacy direct and measured scenarios use 100 assignments, and explicitly exploratory diagnostics use five. Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment responses; here fiscal policy is held fixed. Only direct taxes and benefits respond. Indirect taxes are not shocked, so reduced consumption-tax receipts are absent from the government balance.

4 Results

All results use FRS 2024–25 and period 2026. The bounds table uses 50 balanced assignments; legacy secondary scenarios use 100. Values after ± are assignment standard deviations, not standard errors or confidence intervals. Numerical Monte Carlo SE is stored separately in each JSON artifact. Five-assignment diagnostics are exploratory.

4.1 Derived sector shocks

The deterministic OBR-style lower-anchor stress is £354 million annually. The unit-stress full-schedule amount is £886 million and the EPD unit stress is £693 million. These are scenario inputs generated by Equation (1), not estimated causal effects. The unit-stress sector rates are exactly half those of the former $\varepsilon = 2$ high case and 2.5 times the OBR-style rates.

4.2 Cushioning

Table 1 reports the two polar bounds of the incidence surface: it crosses the external OBR-style and deliberately stronger unit anchors with one common-loss contrast and three duration-equivalent scales. At the 12-month unit stress, wage-cut cushioning exceeds displacement cushioning by 10.0 percentage points (assignment SD 4.7), while its Exchequer cost differs by £107 million. These are conditional statutory comparisons, not confidence intervals for tariff effects. The mixed grid below is the conceptual centre of the design; the bounds make its composition effect transparent.

Assignment-design and record-support audit. At the unit 12-month stress, balanced and Bernoulli integration agree closely: cushioning is 34.0 versus 33.5 per cent and Exchequer cost is £378 versus £377 million. Balancing reduces gross-loss SD to 0.02 of the Bernoulli value and Exchequer SD to 0.18, but it cannot repair sparse survey support. The unit displacement loss has only 6.2 effective record contributions and its largest record supplies 28.7 per cent on average. At

Table 1: Primary common-loss contrast by anchor and duration-equivalent annual stress, 50 assignments

Anchor	Months	Margin	Gross loss (£m)	Exchequer (£m)	Cushioning (%)	Poverty (pp)
OBR	3	Displacement	89 ± 2	35 ± 8	30.9 ± 9.0	0.003
OBR	3	Wage cut	89	48	43.8	0.000
OBR	6	Displacement	177 ± 3	73 ± 13	32.7 ± 7.7	0.009
OBR	6	Wage cut	177	96	43.7	0.000
OBR	12	Displacement	354 ± 4	141 ± 21	32.0 ± 5.7	0.015
OBR	12	Wage cut	354	192	43.6	0.000
UNIT	3	Displacement	222 ± 4	91 ± 15	32.0 ± 6.2	0.008
UNIT	3	Wage cut	221	120	43.6	0.000
UNIT	6	Displacement	442 ± 6	190 ± 24	33.8 ± 5.4	0.016
UNIT	6	Wage cut	443	241	43.6	0.000
UNIT	12	Displacement	885 ± 6	378 ± 37	34.0 ± 4.7	0.036
UNIT	12	Wage cut	886	484	43.9	0.000

Notes: OBR uses export-demand elasticity 0.4; unit uses 1.0. Values after ± are assignment SDs. Displacement uses balanced numerical integration; wage cuts are deterministic apart from the common UC claiming redraw. Three- and six-month rows scale the annual probability of a coherent full-year displaced state; they match expected worker-month loss but do not simulate partial-year unemployment spells. Poverty is the change in the person-weighted fixed-line BHC rate.

the OBR three-month scale these deteriorate to 1.2 and 91.9 per cent. The smallest displacement rows are consequently sensitivity evidence, not precise population estimates. Across leave-each-sector-out reruns for all 23 exposed divisions, the wage-minus-displacement cushioning difference remains positive, ranging from 7.6 to 11.3 percentage points. No single division creates or reverses the primary contrast.

Survey sampling uncertainty. Assignment SDs do not measure sampling variance, so the primary contrast is additionally scored with a household bootstrap of the FRS sample: 999 multinomial resamples of the 16,288 households, applied to per-household contributions from 25 common assignment draws of each unit 12-month margin, so that resampling variation is measured around the draw-averaged statistic. The wage-minus-displacement cushioning difference is 10.7 percentage points with a bootstrap standard error of 1.2 points and a 95 per cent percentile interval of 9.1 to 13.9 points: the interval excludes zero, so the sign of the primary contrast is not an artefact of which households the FRS happened to sample, although its magnitude is estimated imprecisely. Margin-level bootstrap SEs are 1.2 points (displacement cushioning), 0.9 points (wage-cut cushioning), and £43 million and £37 million on the respective Exchequer costs. The FRS research file carries no PSU or stratum identifiers, so households are the resampling unit; the interval approximates the design-based sampling variance without capturing geographic clustering, and it does not incorporate first-stage or parameter uncertainty, which remain scenario dimensions. One convention should be noted: a resample can zero a displacement draw's gross loss, leaving that draw's cushioning ratio undefined; replicates average over the defined draws, and near-zero-gross resamples with unstable ratios are retained. The interval is therefore a record-resampling sensitivity, not evidence that a conditional scenario contrast is statistically identified.

Part of the wage-cut margin's higher measured cushioning reflects a pension-contribution channel rather than the tax–benefit system alone: employee pension and salary-sacrifice contributions are scaled proportionally with the earnings cut, so part of the gross earnings loss is absorbed by reduced retirement saving rather than by current disposable income, while the associated tax relief falls. This is a measurement choice—the HBAI disposable-income concept treats employee pension contributions as an outflow—and it carries a caveat: the implied loss of deferred consumption in retirement is not captured by the cushioning rate.

Mixed and transition paths. The organizing mixed scenario uses a churn-dominated 85/15 displacement–survivor-earnings split, yielding a cushioning rate of 35.9 ± 4.3 per cent and an Exchequer cost of £406 ± 210 million under the full schedule. This split is a transparent scenario choice, not a conversion of a rent-sharing elasticity into an estimated adjustment share. The new transition-equivalent path assigns 70 per cent of each sector's declared loss to workers whose annual earnings and hours embody a three-month re-employment lag before entering a services job at the FRS 28.3 per cent earnings penalty; survivor cuts are rescaled so expected sector loss remains common. It yields 41.4 ± 2.4 per cent cushioning and an Exchequer cost of £461 ± 135 million. The 70/30 share, lag and penalty are organising assumptions bounded by external evidence, not estimated tariff responses. The response surface and lower 14 per cent penalty sensitivity are reported in the supplement.

4.3 External magnitude and outturn checks

The frozen HMRC customs build totals £55.6 billion of 2024 US-bound goods exports before exclusions and maps £52.5 billion to producing divisions. This is reasonably close to the ONS balance-of-payments headline of £59.3 billion, given known conceptual and coverage differences (ONS, 2025). The reference full-schedule calibration predicts a 12.57 per cent aggregate fall across mapped divisions. The observed May 2025–February 2026 customs window falls 12.71 per cent against its two-year same-month baseline. This near equality is not a validation result: the reference calibration was not estimated from that window, and the outturn combines tariff effects with prices, exchange rates, anticipation, composition, rerouting and other demand shocks.

Sector agreement is much weaker. The stored validation artifact reports a rank correlation of 0.00 between the full-schedule predicted and realised falls among the ten largest exposed divisions, and a correlation of 0.58 between full-schedule and observed downside shock rates across all mapped divisions. Machinery and motor vehicles decline broadly in the expected direction, while aerospace, electrical equipment and basic metals do not. The aggregate match can therefore be coincidental and must not be used to validate Equation (1). As a separate policy benchmark, the OBR's March 2025 scenario uses a demand elasticity of 0.4, implying an 8 per cent export-demand fall after a 20 percentage-point tariff and a direct export-demand effect just under 0.2 per cent of GDP (OBR, 2025). The paper's $\varepsilon = 1$ reference is 2.5 times that elasticity and additionally assumes full wage-bill incidence; it is a deliberately strong stress, not a central forecast.

Product–destination panel. Across 1194 HS4 products, the capped value-weighted specification estimates a -31.9 per cent change in exports to the US relative to the same products sent to the three comparison destinations (95 per cent interval -44.8 to -16.0 per cent, $p =< 0.001$). Results have similar signs against each destination, and May 2019–23 pseudo-policy estimates are individually insignificant. This is not robust evidence of a common product effect: equal product weights give -4.9 per cent ($p = 0.293$), and restricting the sample to 2023 onward gives

-17.5 per cent ($p = 0.080$). The estimated differential trend is itself borderline ($p = 0.057$). Most importantly, broad steel and auto chapters fall -20.0 per cent ($p = 0.290$), compared with -32.9 per cent for other products ($p = 0.001$): the tariff-intensity ordering fails. Section 7 displays the monthly gap. The panel supports a post-policy decline concentrated in large US export products, but it cannot attribute that decline to tariffs or validate the tariff-rate mapping.

Why this failure does not undermine the paper's estimand deserves a direct statement rather than an implication. The falsification tests the *bridge*—the claim that sector-level tariff rates predict sector-level export falls—and the bridge fails at sector rank. But the primary estimand never conditions on that claim: it asks how the tax–benefit system treats a *declared* sector wage-bill loss delivered through two adjustment margins, and the declared loss is identical across margins by construction. A different sector composition of the same aggregate loss changes which households are exposed, not the statutory asymmetry between intensive and extensive relief; the leave-one-division-out audit (all 23 divisions, contrast range 7.6–11.3 points) and the three LFS-shaped selection models test exactly this and the contrast survives every rerun. The failed falsification therefore bounds the paper's claims—the sector calibration must not be read as measured tariff incidence—without touching the cushioning comparison, which is the result the paper is submitted on. What the failure *does* rule out is any future reading of Equation (1) as an estimated first stage; that reading is disclaimed throughout.

Longitudinal worker benchmark. The five-quarter LFS contains 177 wave-1 manufacturing employees after the required population filters, compared with 1102 employed manufacturing FRS receivers. Its BRES-composition-adjusted manufacturing targets are an 8.96 per cent wave-1-to-wave-5 job-exit rate and 5.87 per cent mean log wage growth among valid endpoint-pay observations. Table 2 compares receiver distributions. Calibration deliberately makes the aggregate exit means identical, so the comparison tests incidence shape rather than level. The QRF has a much wider risk distribution and agrees weakly with the primary cells: person-level correlations are only 0.06 for exit risk and 0.08 for predicted wage change. The longitudinal evidence therefore does not support replacing transparent cells with forest-generated tariff incidence. It instead quantifies model uncertainty that the central uniform assignment omits. Regularisation does not eliminate all forest tails: the maximum calibrated exit risk is 62.6 per cent and the predicted log-wage-change range is -45.7 to 72.0 per cent. These are benchmark predictions, not plausible confidence bounds.

Table 2: Calibrated FRS job-exit-risk distributions

Estimator	Weighted mean	P10	Median	P90
Calibrated SIC–sex–age cells	8.96	7.15	8.42	11.88
Calibrated income terciles	8.96	3.88	8.91	13.77
Regularised QRF benchmark	8.96	1.22	5.42	19.62

Notes: Entries are percentages among matched employed manufacturing FRS receivers. All means equal the same directly measured LFS target by construction. Quantiles are survey-weighted. QRF denotes the regularised quantile/random-forest benchmark described in Section 3.5; it predicts baseline transitions, not causal tariff effects.

Fiscal sensitivity to worker-risk shape. Table 3 uses each calibrated risk model to vary selection within SIC divisions while holding every division's expected weighted wage-bill loss at the original shock. The cell model nearly reproduces uniform assignment. Across all three

LFS-shaped models, mean cushioning ranges from 36.0 to 37.4 per cent, Exchequer cost from £384 million to £425 million, and measured poverty change from 0.024 to 0.028 percentage points. This model spread is smaller than the assignment SD within any specification. Weighted displaced headcount changes because calibration preserves the wage bill, not a headcount quota. The QRF mean gross loss is £847 million rather than the £886 million expectation, but its £397 million assignment SD implies a numerical MC SE of about £40 million, so this difference is not evidence of a different aggregate target.

Table 3: Full-schedule fiscal sensitivity to within-SIC worker selection

Selection model	Workers (000s)	Gross loss (£m)	Exchequer (£m)	Cushioning (%)	Poverty (pp)
Uniform central assignment	17.2 ± 6.9	882 ± 418	387 ± 213	34.1 ± 5.2	0.032 ± 0.022
LFS calibrated cells	17.2 ± 6.9	881 ± 412	412 ± 223	36.9 ± 5.0	0.028 ± 0.019
LFS income terciles	15.2 ± 6.5	887 ± 467	425 ± 258	37.4 ± 6.0	0.025 ± 0.017
LFS regularised QRF	18.1 ± 5.9	847 ± 397	384 ± 212	36.0 ± 5.7	0.024 ± 0.016

Notes: Means ± assignment SD over 100 common assignments. Within each SIC division, individual probabilities are rescaled so the earnings-weighted mean equals the declared sector shock. Poverty is relative BHC against the frozen baseline line. LFS risks are predictive baseline-transition scores, not tariff-specific causal probabilities.

The realised displacement output is also consistent with the allocation design but too sparse for precise empirical incidence: the expected 9.7 selected FRS records per assignment produce 17.2 thousand represented displaced workers and £882 ± 418 million of gross loss. The gross-loss SD is nearly half its mean, and Exchequer-cost SD exceeds half its mean. The 100-assignment average is numerically more stable than an individual assignment, but it remains conditional on the observed 1,018 exposed records and the uniform within-division selection rule.

Take-up sensitivity. The regenerated grid applies the same post-shock-entitlement claiming rule to all margins. It uses only five assignments per cell and is exploratory. It shows that displacement responds more visibly to the assumed claiming probability, while wage-cut results move little because relatively few wage-cut households change UC entitlement. No grid difference is treated as statistically identified.

4.4 Mechanism

The regenerated decomposition and gate audit are also five-assignment diagnostics. Income-tax relief is larger on the wage-cut margin than on displacement; UC is more important after displacement. The plain FRS input contains no usable household-capital measure, so the model cannot establish the empirical importance of UC's capital test. New-style JSA is not activated by the imposed transition. These omissions make the gate audit a description of the implemented model, not the full UK out-of-work system.

4.5 Distributional incidence

Under full displacement, relative BHC poverty rises by 0.032 ± 0.022 percentage points. Under wage cuts it is unchanged at displayed precision. These national changes are small relative to omitted uncertainty and are not headline estimates. They can coexist with large selected-household losses because exposure is limited and concentrated above the poverty line. Decile profiles and assignment dispersion are reported in Section 7.

4.6 Fiscal incidence

Annual Exchequer costs are reported in Table 1. The measure is the aggregate simulated government-balance change and is broader than the household cushioning identity. Employer National Insurance contributions and the tax consequences of pension and salary-sacrifice changes respond to the shock but sit outside employee disposable income. Section 7 provides the formal reconciliation and draw distributions.

4.7 EPD counterfactual

On paired seeds, full minus EPD displacement differs by 3.7 ± 2.8 thousand workers and £90 ± 98 million of Exchequer cost. These are secondary scenario contrasts, not causal deal estimates; detailed gross-loss, poverty and graphical results are in the Section 7.

5 Policy

5.1 Economic Prosperity Deal

Within the maintained direct-channel scenarios, the paired full-minus-EPD displacement contrast is 3.7 ± 2.8 thousand workers, £204 ± 192 million of gross earnings and £90 ± 98 million of annual Exchequer cost. These SDs describe assignment dispersion and the contrasts are not causal estimates of the deal. Autos drive much of the difference; the scenario applies the in-quota rate to fixed exposure without modelling quota allocation or future above-quota trade.

5.2 Pharmaceutical channel

The EPD scenario sets the pharmaceutical employment shock to zero, but this does not make the exemption free. Its VPAG and NHS pricing concessions sit outside the labour-income microsimulation. A complete policy appraisal would cost those health-budget and price effects alongside the direct tax–benefit scenario.

5.3 Cushioning policy

The simulations support a limited conclusion: extensive-margin protection depends more on means-tested eligibility and claiming than the income-tax response to an intensive earnings loss. The five-assignment take-up grid is exploratory and does not quantify claimant behaviour precisely. Automatic enrolment after redundancy, contribution-based unemployment insurance and wage insurance are therefore candidate counterfactuals for future analysis, not policy recommendations established by the present results. Their costs and incidence require dedicated simulations and financing assumptions.

The repository also supplies reusable interfaces for the next policy exercises. Observable household and worker characteristics can tilt shock probabilities while preserving each sector's aggregate earnings loss, and a separate price-channel interface converts expenditure-weighted price changes into real disposable income. Neither interface supplies causal coefficients: the former requires externally estimated heterogeneity and the latter requires household expenditure data. They are extensions to populate and validate, not additional headline estimates in this paper.

6 Discussion and conclusion

The central finding is that the UK tax–benefit system responds differently when a common aggregate earnings loss is distributed through alternative labour-market margins. Across two loss anchors and three duration-equivalent scales, the wage-cut and displacement bounds differ by 9.8–12.8 percentage points; at the unit twelve-month scale the difference is 10.0 points, the wage-cut Exchequer cost is £107 million higher, and the household bootstrap places the difference at 10.7 points with a 95 per cent interval of 9.1–13.9 points in the record-resampling interval. The mechanism is institutional: proportional cuts receive tax relief near marginal rates, while whole-job loss receives relief nearer average rates and relies more on UC—relief that is gated by means tests, capital rules and claiming behaviour. Part of the wage-cut difference is lower pension saving under the HBAI cash-income measure, not current tax-benefit insurance. The contrast is pre-specified and conditional: it prices a statutory asymmetry, not the tariffs' total causal incidence.

Balanced integration makes the common aggregate loss comparable assignment by assignment and closely reproduces the Bernoulli mean at unit scale. It does not overcome sparse information: the unit displacement loss has only 6.2 effective record contributions, and support is weaker at the smallest OBR-duration cells. Assignment SDs describe scenario-allocation sensitivity; sampling uncertainty is scored separately by the household bootstrap, which is a sensitivity interval rather than a design-based confidence interval. The sign also survives all 23 leave-one-division-out reruns, with a cushioning difference of 7.6–11.3 points. This establishes sector robustness within the imposed bridge, not causal robustness of that bridge.

Calibrated longitudinal-LFS allocation sensitivities leave the central extensive-margin conclusion broadly unchanged: cushioning spans 36.0–37.4 per cent across cell, earnings-band and regularised-QRF risk shapes, compared with 36.8 per cent under uniform assignment. This robustness is conditional and limited. The imputation models agree weakly person by person, and the available LFS manufacturing target has only 177 donors; stability of an aggregate fiscal ratio is not evidence that tariff-specific worker selection has been identified.

National poverty changes remain near zero because the exposed population is small and often above the poverty line. They are not precise population estimates and do not imply that affected households' losses are small. EPD, inactivity, reallocation and demographic results are secondary or supplementary rather than competing headline estimands.

The common-loss transition-equivalent path makes the preferred economic interpretation more realistic than either polar bound: some workers receive an annualized re-employment penalty, some re-enter services at lower earnings, and survivors absorb the residual loss. Under the organizing 70/30 transition allocation, the simulated cushioning is 41.4 per cent and the Exchequer cost is £461 million. These are conditional results: the annualized lag is encoded in earnings and hours rather than a monthly unemployment-benefit spell; the transition share, three-month lag and 28.3 per cent services penalty are scenario inputs, and the latter is a cross-sectional FRS destination gap that also captures hours and part-time composition. The response surface, including 50/50 and 85/15 allocations and the 14 per cent penalty, is therefore more informative than any single preferred point.

The main limitation is structural: the model starts at the end of the production-side transmission mechanism. It does not estimate how the tariff changes border prices, export quantities, domestic output, value added, imported-input costs, firm productivity, investment, profits or labour demand. It assigns calibrated exposure directly to the wage bill and then assumes how that loss is divided among wages, employment, inactivity and reallocation. The results therefore answer a conditional fiscal-incidence question—what statutory taxes and transfers do after specified labour-income changes—not what the tariff does to the economy. Assignment is also independent within broad SIC divisions. The new longitudinal LFS exercise shows that plausible cell and forest incidence shapes agree only weakly even after matching aggregate targets; with just 177 manufacturing donors it cannot identify tariff-specific worker selection. The inactivity scenario assumes WCA passage for selected older workers; New Style JSA and within-year timing are absent; household labour-supply responses are suppressed; and indirect taxes, exchange rates, market switching, retaliation and general equilibrium are outside the central model. The supply-chain calculation is an accounting sensitivity, not a lower bound, and cannot be mechanically combined with local-employment multipliers. Constituency results in Section 7 are illustrative because local industry is not observed at the required level.

The companion UK AI study suggests useful extensions to the analysis architecture: separate the aggregate shock size from its incidence pattern as factorial scenario axes; benchmark exposure against external employment data; distinguish evidence-guided from measured-incidence scenarios; evaluate policy counterfactuals on common draws; and generate paper values from a declared build manifest. These extensions improve transparency and robustness, but they do not replace the missing trade-to-production first stage. The priority for a substantive extension is therefore a product- or firm-level model linking tariffs to prices, quantities, output, value added and productivity before any household microsimulation is run.

The defensible conclusion is therefore modest. For a given direct earnings loss, the adjustment margin materially changes the mix of tax relief, benefit support, poverty and fiscal cost. The polar margins bracket a range of adjustment assumptions, and the 15/85 mixed result is a literature-disciplined sensitivity rather than an estimated decomposition. The close aggregate match between the reference and observed export falls does not validate the bridge because sector agreement is weak. Quantifying the realised tariff effect requires causal trade and linked employer–worker evidence beyond this microsimulation.

7 Additional Results and Diagnostics

US exports use HMRC uktradeinfo OTS/RTS data for 2024, mapped from SITC to SIC 2007 divisions and divided by the latest non-suppressed ONS Annual Business Survey turnover. Of a £55,600 million customs-basis total, £52,500 million maps to divisions after documented exclusions, chiefly non-monetary gold and precious metals. The measured window uses May 2025–February 2026 against the mean of the same calendar months one and two years earlier. Its 12.7 per cent aggregate fall is descriptive, not causal.

8 Shock mechanics

Each employee in exposed division j is independently selected with probability s_j. Weighted displaced counts and earnings therefore meet targets in expectation rather than exactly. Selected workers have employment income, hours, pension contributions and statutory pay zeroed. Wage cuts remove s_j of each exposed worker's earnings deterministically. Reallocation uses the same selected workers, assigns a services destination, and applies the calibrated earnings penalty.

UC claiming is redrawn after every margin for benefit units whose circumstances changed and that are entitled post-shock. The LCWRA override sets the benefit-unit element to the maximum of its stored baseline value and one statutory annual element, preventing duplicate payment.

9 Monte Carlo and measures

Central direct and measured scenarios use 100 assignments. Every displayed ± value is an assignment SD, not a standard error or confidence interval. The machine-readable artifacts additionally store numerical Monte Carlo standard error as assignment SD divided by $\sqrt{n}$; it measures precision of the simulated mean only. Survey-design, tariff-response and model-parameter uncertainty are not estimated. Five-assignment take-up, mechanism, demographic, poverty-gap, reallocation, duration and sensitivity-grid exercises, and the ten-assignment supply-chain exercise, are exploratory and are excluded from the headline evidence. The mixed-adjustment scenario surface likewise uses five common assignments per cell. Its machine-readable cells and draws are in `results/scenario_testing.json` and `results/scenario_testing.csv`.

Reported relative BHC poverty changes hold the poverty line frozen at 60 per cent of the baseline equivalised median; absolute BHC poverty likewise uses 60 per cent of the baseline equivalised median as a fixed line. The cushioning rate is $1 - \Delta Y/\Delta E$, where ΔE is gross employment-earnings loss and ΔY is household disposable-income loss.

Reconciling cushioning and the Exchequer effect. The Exchequer effect is the change in the aggregate simulated government balance, not the household-side transfer implied by the cushioning identity. The identity implies a household-side offset of $(1 - \Delta Y/\Delta E) \times \Delta E = \Delta E - \Delta Y$: for the full-schedule wage cut, $0.439 \times £886\text{m} \approx £389$ million, and for displacement $0.368 \times £882\text{m} \approx £325$ million, against reported Exchequer costs of £484 million and £387 million. The wedge is revenue lost outside employee disposable income: employer National Insurance contributions fall with the employee wage bill under every margin, and for displaced workers the zeroing of employee pension contributions and salary-sacrifice arrangements changes taxable pay relative to the gross-earnings accounting in ΔE. The government balance also absorbs any other simulated fiscal flows that respond to the shock. Cushion × gross therefore understates the Exchequer cost by roughly £87–95 million in the central full-schedule scenarios, and the two statistics answer different questions: the cushioning rate measures household income protection per pound of employee earnings lost, while the Exchequer effect measures the total first-round cost to the public finances.

10 Elasticity and pass-through grid

The regenerated grid spans $\varepsilon \in \{0.4, 1, 2, 3\}$ and $\pi \in \{0.5, 0.75, 1\}$. Because only five displacement assignments are used per cell, it is an exploratory scaling diagnostic. Equal products $\varepsilon\pi$ generate equal deterministic wage-cut shocks; Bernoulli displacement outcomes vary across assignments. The grid is not an uncertainty interval.

11 Factorial adjustment scenarios

Figure 1 separates the aggregate export-demand calibration from the assumed composition of labour-market adjustment. Each cell uses the full tariff schedule and five common assignments.

The wage-cut endpoint is stable: cushioning is 40.6–41.3 per cent over the four calibrations. Moving towards displacement generally lowers cushioning, but the small-shock cells are especially noisy because a few high-weight FRS records dominate five Bernoulli assignments. These cells are stress scenarios, not estimates or confidence regions. The qualitative result is that fiscal cost scales mainly with the imposed gross shock, while the adjustment mix changes the share absorbed by taxes and transfers.

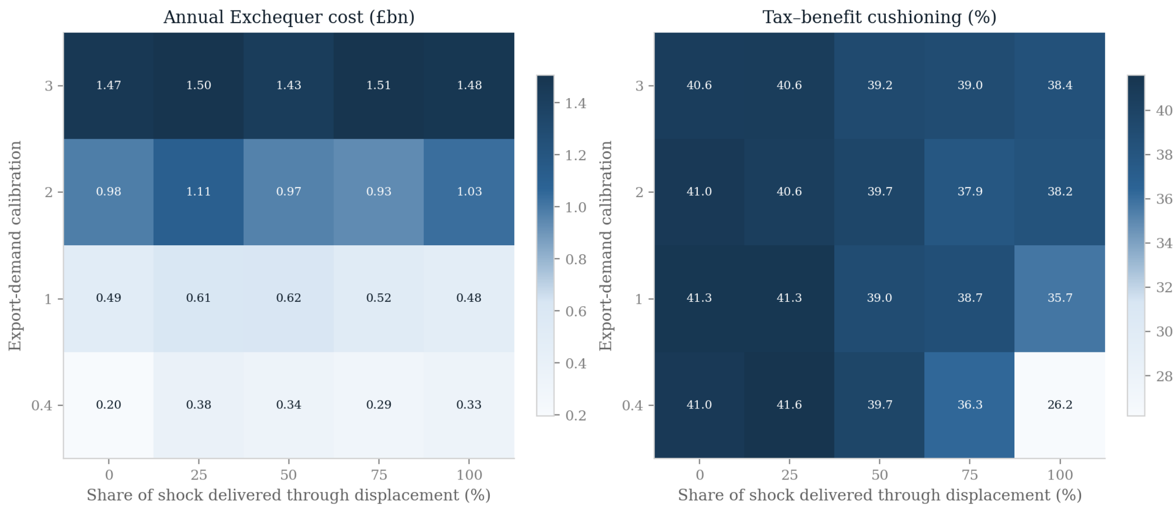

Figure 1: Factorial scenario surface: export-demand calibration by the share of expected sector loss delivered through displacement rather than survivor wage cuts. Cell values are means over five common assignments. Assignment noise, survey uncertainty and uncertainty in the production-side transmission are not combined.

12 Reallocation sensitivity

The reallocation exercise uses five paired assignments and is exploratory. Under the full schedule, the central 28.3 per cent penalty produces a mean gross loss of £264 million, Exchequer cost of £149 $\pm$ 131 million, cushioning of 39.7 $\pm$ 3.4 per cent, and zero measured BHC-poverty change. With a three-month lag, gross loss rises to £432 million and the Exchequer cost to £245 $\pm$ 216 million. The 14 per cent penalty gives gross loss of £131 million and Exchequer cost of £75 $\pm$ 64 million. These pound levels are not commensurable with the economy-wide wage-cut scenario because only selected movers' penalty is removed.

13 Observed-outturn stress scenario

The descriptive May 2025–February 2026 export window falls 12.7 per cent against its two-year same-month baseline. This is not a causal tariff estimate: it includes prices, exchange rates, front-running, rerouting and other demand conditions.

The downside displacement scenario affects 17.3 thousand weighted workers on average, is cushioned by 34.0 $\pm$ 5.0 per cent and costs the Exchequer £394 $\pm$ 218 million. The wage-cut scenario is cushioned by 44.1 per cent and costs £517 million. This is an observed-outturn downside stress case, not a tariff-caused or preferred estimate; the validation artifact also reports signed net exposure before expanding divisions are clipped.

14 HMRC destination-panel diagnostic

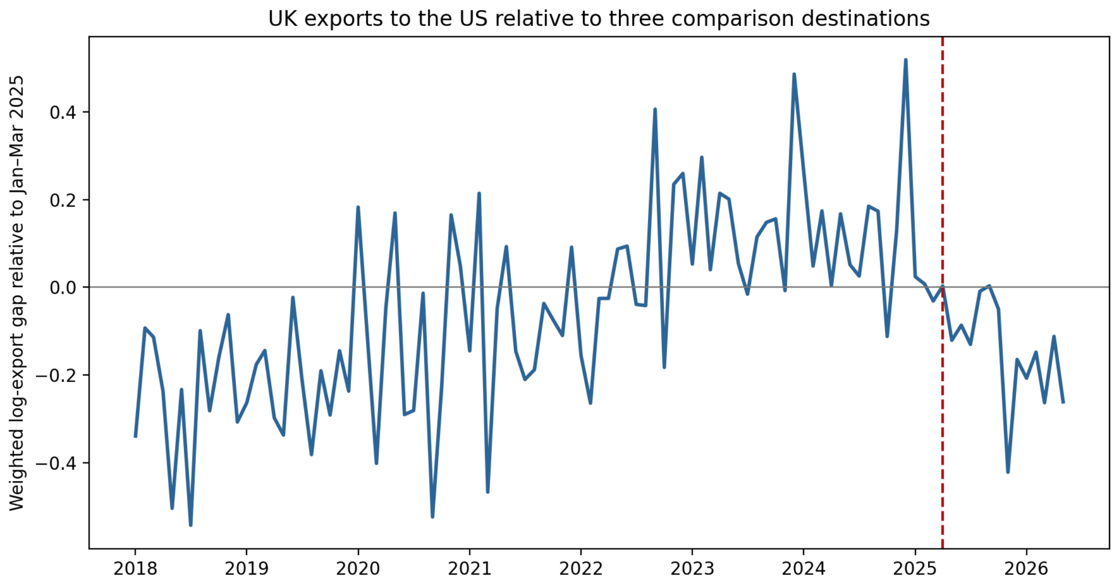

Figure 2: Monthly capped-value-weighted gap between UK HS4 exports to the United States and mean exports of the same products to Canada, Japan and Australia, normalised to January–March 2025. The dashed line marks April 2025, which is omitted from the regression as an announcement and anticipation month. The series is descriptive and has no causal confidence band.

The machine-readable specification, clustered standard errors, individual destination checks, sample-start sensitivities, tariff-intensity groups and May 2019–23 pseudo-policy estimates are stored in the event-study JSON artifact; the plotted observations are stored in its companion monthly CSV. Pre-policy value weights are capped at the 99th percentile, giving an effective product count of 107 despite 1194 included HS4 products. Uncapped and equal-weight estimates disclose the remaining concentration sensitivity.

15 Demographic diagnostic

The gender, age and household-type breakdown uses five Bernoulli assignments and is exploratory. Selected exposed-sector workers are disproportionately men and concentrated in prime and older working ages, but individual subgroup estimates are too assignment-sensitive for precise rankings. The demographic artifact is therefore a diagnostic rather than an inferential table.

16 Duration scope

The submission design reports three-, six- and twelve-month *duration-equivalent annual stresses*. It scales the probability of a coherent full-year displaced state rather than combining partial-year earnings with full-year unemployment status. These rows preserve expected worker-month loss but are not unemployment-spell simulations. Actual within-year tax, UC, waiting-period and re-employment timing requires a monthly model and is not claimed here.

17 Secondary distributional and fiscal figures

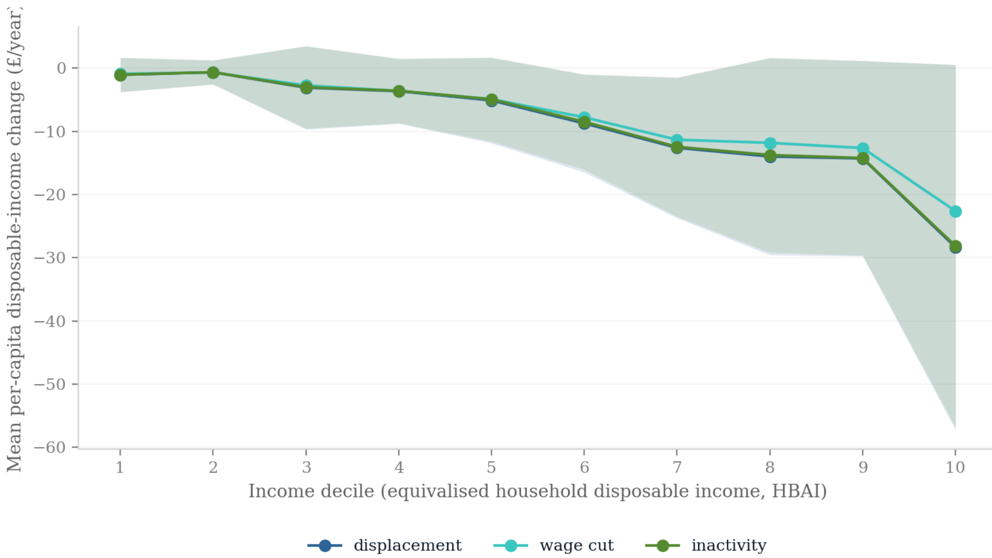

Figure 3: Per-capita disposable-income change by baseline income decile in the legacy full-schedule scenarios. Shading is assignment dispersion.

18 Supply-chain sensitivity

The input–output scenario adds upstream requirements using the ONS domestic Leontief inverse. The accounting shock has a 1.95 direct-plus-upstream factor. It is not a lower bound and is not multiplied by local-employment multipliers. In the exploratory ten-assignment microsimulation, displacement affects 29,700 weighted workers on average, costs the Exchequer £680 $\pm$ 310 million and raises relative BHC poverty by 0.028 percentage points. The wage-cut variant costs £960 million and leaves measured relative poverty unchanged. These figures are accounting sensitivities, not all-channel causal estimates.

19 Constituency illustration

The enhanced-FRS constituency layer uses imputed industries rather than observed local industry structure. It is averaged over 20 paired worker-assignment draws (seeds 0–19), replacing the

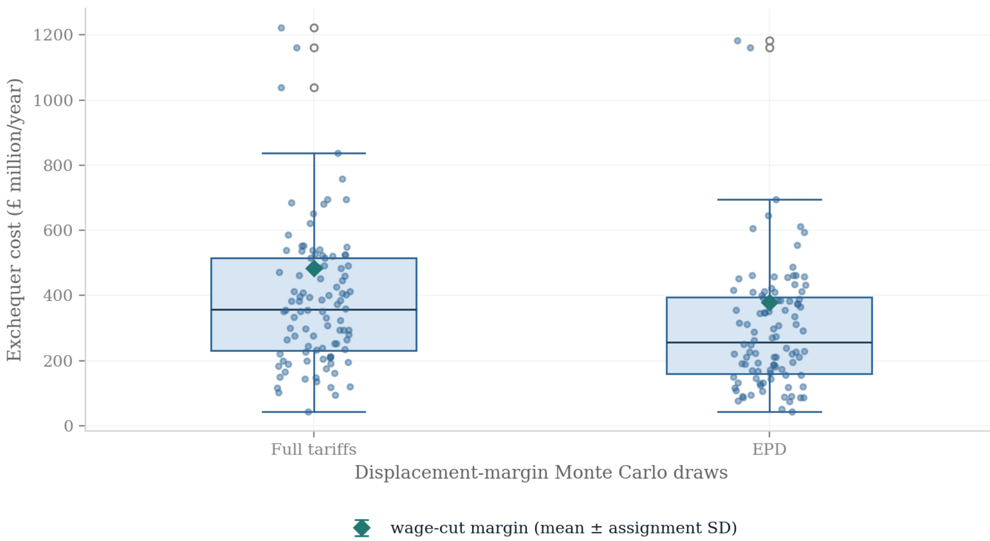

Figure 4: Annual Exchequer cost across legacy regenerated assignments.

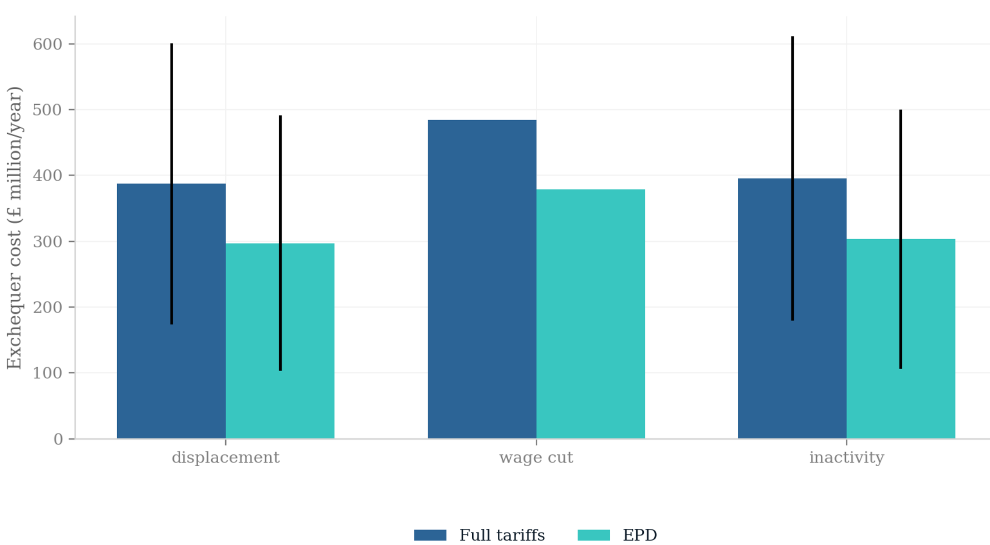

Figure 5: Full-schedule and EPD scenario comparison. Error bars are assignment standard deviations, not causal confidence intervals.

unstable single-seed surface used in an earlier draft. The mapping now divides each household's disposable-income change by household size before constructing constituency means. The CSV reports both the draw mean and assignment-draw standard deviation. Figure 6 shows the resulting synthetic surface. Because the constituency weights contain no observed local industry mix and the SIC imputation is held fixed, these remain synthetic demographic and income projections, not constituency exposure estimates; individual seat rankings are therefore deliberately not reported.

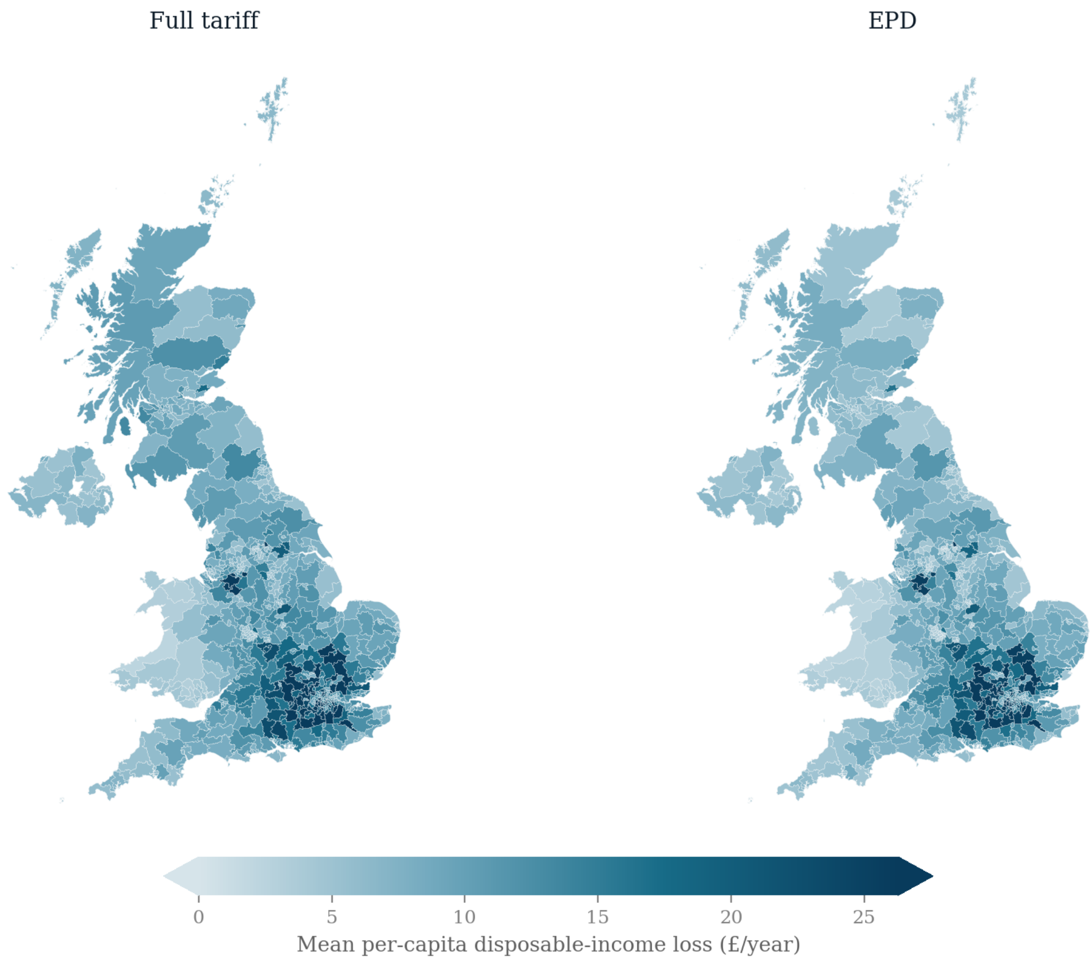

Figure 6: Synthetic constituency illustration of annual per-capita disposable-income loss, averaged over 20 paired worker-assignment draws. A shared sequential scale is clipped at the 95th percentile. The map uses imputed SIC and no observed constituency industry mix; it is not a map of local tariff exposure.

References

Office for National Statistics (2025). *Labour Force Survey Five-Quarter Longitudinal Dataset, July 2024–September 2025*. UK Data Service, Study Number 9490. doi:10.5255/UKDA-SN-9490-1.

HM Revenue and Customs (2026). *UK overseas trade in goods statistics: detailed commodity by country data*. HMRC Online Trade Statistics API, monthly observations through May 2026. https://www.uktradeinfo.com/.

ABPI (2025). UK–US agreement on pharmaceutical tariffs and pricing. Association of the British Pharmaceutical Industry, London, December 2025.

Acemoglu, D., Akcigit, U. and Kerr, W. (2016). Networks and the macroeconomy: An empirical exploration. *NBER Macroeconomics Annual*, 30(1), 273–335.

Acosta, M. and Cox, L. (2024). The regressive nature of the US tariff code: Origins and implications. Working paper, February 2024 (conditionally accepted, *The Quarterly Journal of Economics*).

Adão, R., Arkolakis, C. and Esposito, F. (2023). General equilibrium effects in space: Theory and measurement. NBER Working Paper No. 25544, National Bureau of Economic Research, Cambridge, MA.

Amiti, M., Redding, S. J. and Weinstein, D. E. (2019). The impact of the 2018 tariffs on prices and welfare. *Journal of Economic Perspectives*, 33(4), 187–210.

Aragón, F. M., Rud, J. P. and Toews, G. (2018). Resource shocks, employment, and gender: Evidence from the collapse of the UK coal industry. *Labour Economics*, 52, 54–67.

Artuc, E., Porto, G. and Rijkers, B. (2021). Household impacts of tariffs: Data and results from agricultural trade protection. *The World Bank Economic Review*, 35(3), 563–585.

Auerbach, A. J. and Feenberg, D. (2000). The significance of federal taxes as automatic stabilizers. *Journal of Economic Perspectives*, 14(3), 37–56.

Atkin, D., Bernadac, V., Donaldson, D., Garg, T. and Huneeus, F. (2025). The incidence of distortions. Working paper, December 2025 draft.

Auray, S., Devereux, M. B. and Eyquem, A. (2020). Trade wars, currency wars. NBER Working Paper No. 27460, National Bureau of Economic Research, Cambridge, MA.

Autor, D. H., Dorn, D. and Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review*, 103(6), 2121–2168.

Autor, D. H., Dorn, D., Hanson, G. H. and Song, J. (2014). Trade adjustment: Worker-level evidence. *The Quarterly Journal of Economics*, 129(4), 1799–1860.

Autor, D. H., Dorn, D. and Hanson, G. H. (2021). On the persistence of the China shock. *Brookings Papers on Economic Activity*, Fall 2021, 381–447.

Bank of England (2025). *Monetary Policy Report*, May, August and November 2025 issues. Bank of England, London.

Barrot, J.-N. and Sauvagnat, J. (2016). Input specificity and the propagation of idiosyncratic shocks in production networks. *The Quarterly Journal of Economics*, 131(3), 1543–1592.

Beatty, C. and Fothergill, S. (2017). The impact on welfare and public finances of job loss in industrial Britain. *Regional Studies, Regional Science*, 4(1), 161–180.

Beatty, C., Fothergill, S. and Powell, R. (2007). Twenty years on: Has the economy of the UK coalfields recovered? *Environment and Planning A*, 39(7), 1654–1675.

Beatty, C. and Fothergill, S. (2023). The persistence of hidden unemployment among incapacity claimants in large parts of Britain. *Local Economy*, doi:10.1177/02690942231184815.

Blanchard, E. J., Bown, C. P. and Chor, D. (2019). Did Trump's trade war impact the 2018 election? NBER Working Paper No. 26434, National Bureau of Economic Research, Cambridge, MA.

Bloom, N., Bunn, P., Chen, S., Mizen, P., Smietanka, P. and Thwaites, G. (2019). The impact of Brexit on UK firms. NBER Working Paper No. 26218, National Bureau of Economic Research, Cambridge, MA.

Borusyak, K. and Jaravel, X. (2021). The distributional effects of trade: Theory and evidence from the United States. NBER Working Paper No. 28957, National Bureau of Economic Research, Cambridge, MA.

Bourguignon, F., Bussolo, M. and Pereira da Silva, L. A. (eds.) (2008). *The Impact of Macroeconomic Policies on Poverty and Income Distribution: Macro-Micro Evaluation Techniques and Tools*. Palgrave Macmillan and World Bank, Washington, DC.

Brewer, M. and Tasseva, I. V. (2021). Did the UK policy response to Covid-19 protect household incomes? *The Journal of Economic Inequality*, 19(3), 433–458.

Breinlich, H., Leromain, E., Novy, D. and Sampson, T. (2022). The Brexit vote, inflation and UK living standards. *International Economic Review*, 63(1), 63–93.

The Budget Lab (2025). Where we stand: The fiscal, economic, and distributional effects of all U.S. tariffs enacted in 2025. The Budget Lab at Yale, New Haven, CT.

Caliendo, L., Dvorkin, M. and Parro, F. (2019). Trade and labor market dynamics: General equilibrium analysis of the China trade shock. *Econometrica*, 87(3), 741–835.

Card, D., Cardoso, A. R., Heining, J. and Kline, P. (2018). Firms and labor market inequality: Evidence and some theory. *Journal of Labor Economics*, 36(S1), S13–S70.

Cavallo, A., Gopinath, G., Neiman, B. and Tang, J. (2021). Tariff pass-through at the border and at the store: Evidence from US trade policy. *American Economic Review: Insights*, 3(1), 19–34.

Cavallo, A., Llamas, P. and Vazquez, F. (2025). Tracking the short-run price impact of U.S. tariffs. NBER Working Paper No. 34496, National Bureau of Economic Research, Cambridge, MA.

Centre for Economic Performance (2025). UK firms and the 2025 US tariffs: Evidence from the Decision Maker Panel. VoxEU column, Centre for Economic Policy Research, and CEP "in brief" 11639, Centre for Economic Performance, LSE, London. https://cepr.org/voxeu/columns/impact-2025-us-tariff-announcements-uk-firms.

Centre for Cities (2025). The uneven storm: How US tariffs hit UK cities differently. Centre for Cities, London.

City-REDI (2025). The regional impact of US automotive tariffs on the UK economy. City-REDI, University of Birmingham.

Corsetti, G., Lloyd, S. and Ostry, D. (2025). Trading blows: The exchange-rate response to tariffs and retaliations. Staff Working Paper No. 1,139, Bank of England, London.

Dauth, W., Findeisen, S. and Suedekum, J. (2021). Adjusting to globalization in Germany. *Journal of Labor Economics*, 39(1), 263–302.

De Lyon, J. and Pessoa, J. P. (2021). Worker and firm responses to trade shocks: The UK–China case. *European Economic Review*, 133, 103678.

Department for International Trade (2021). *Trade Modelling Review Expert Panel: Report and Recommendations*. Department for International Trade, London.

den Besten, T. and Känzig, D. R. (2025). The macroeconomic effects of tariffs: Evidence from U.S. historical data. NBER Working Paper No. 34852, National Bureau of Economic Research, Cambridge, MA.

Dhingra, S. (2025). Tariffed with the same brush. Speech, Bank of England, October 2025.

Dhingra, S., Ottaviano, G., Sampson, T. and Van Reenen, J. (2017). The costs and benefits of leaving the EU: Trade effects. *Economic Policy*, 32(92), 651–705.

Dix-Carneiro, R. (2014). Trade liberalization and labor market dynamics. *Econometrica*, 82(3), 825–885.

Dix-Carneiro, R. and Kovak, B. K. (2017). Trade liberalization and regional dynamics. *American Economic Review*, 107(10), 2908–2946.

Dolls, M., Fuest, C. and Peichl, A. (2012). Automatic stabilizers and economic crisis: US vs. Europe. *Journal of Public Economics*, 96(3–4), 279–294.

Dorn, D. and Levell, P. (2024). Labour market impacts of the China shock: Why the tide of globalisation did not lift all boats. *Labour Economics*, 91, 102629.

Department for Work and Pensions (2024). Income-related benefits: Estimates of take-up. Official statistics series, DWP, London.

Fajgelbaum, P. D. and Khandelwal, A. K. (2016). Measuring the unequal gains from trade. *The Quarterly Journal of Economics*, 131(3), 1113–1180.

Fajgelbaum, P. D. and Khandelwal, A. K. (2022). The economic impacts of the US–China trade war. *Annual Review of Economics*, 14, 205–228.

Fajgelbaum, P. D. and Khandelwal, A. K. (2026). Tariffs in 2025: Short-run impacts on the U.S. economy. *Brookings Papers on Economic Activity*, Spring 2026 conference draft.

Fajgelbaum, P. D., Goldberg, P. K., Kennedy, P. J. and Khandelwal, A. K. (2020). The return to protectionism. *The Quarterly Journal of Economics*, 135(1), 1–55.

Fetzer, T. (2019). Did austerity cause Brexit? *American Economic Review*, 109(11), 3849–3886.

Fetzer, T. and Wang, S. (2024). Measuring the regional economic cost of Brexit: Evidence up to 2019. CAGE Working Paper No. 1280, University of Warwick.

Flaaen, A. and Pierce, J. R. (2019). Disentangling the effects of the 2018–2019 tariffs on a globally connected U.S. manufacturing sector. Finance and Economics Discussion Series 2019-086, Board of Governors of the Federal Reserve System, Washington, DC.

Galle, S., Rodríguez-Clare, A. and Yi, M. (2023). Slicing the pie: Quantifying the aggregate and distributional effects of trade. *The Review of Economic Studies*, 90(1), 331–375.

Garin, A. and Silvério, F. (2024). How responsive are wages to firm-specific changes in labour demand? Evidence from idiosyncratic export demand shocks. *The Review of Economic Studies*, 91(3), 1671–1710.

Gathmann, C., Helm, I. and Schönberg, U. (2020). Spillover effects of mass layoffs. *Journal of the European Economic Association*, 18(1), 427–468.

Hummels, D., Jørgensen, R., Munch, J. and Xiang, C. (2014). The wage effects of offshoring: Evidence from Danish matched worker–firm data. *American Economic Review*, 104(6), 1597–1629.

Hyman, B., Kovak, B. K. and Leive, A. (2024). Wage insurance for displaced workers. NBER Working Paper No. 32464, National Bureau of Economic Research, Cambridge, MA.

Institute for Fiscal Studies (2024). Health-related benefit claims post-pandemic: UK trends and global context. IFS Report, Institute for Fiscal Studies, London, September 2024.

International Monetary Fund (2025). *World Economic Outlook: Global Economy in Flux, Prospects Remain Dim*. IMF, Washington, DC, October 2025.

Ignatenko, A., Lashkaripour, A., Macedoni, L. and Simonovska, I. (2025). Making America great again? The economic impacts of Liberation Day tariffs. *Journal of International Economics*, 157, 104138.

Irastorza-Fadrique, A., Levell, P. and Parey, M. (2025). Household responses to trade shocks. *Journal of International Economics*, forthcoming (IFS Working Paper 24/52).

Jeanne, O. and Son, J. (2024). To what extent are tariffs offset by exchange rates? *Journal of International Money and Finance*, 142.

Kalemli-Özcan, Ş., Soylu, C. and Yildirim, M. A. (2026). Global trade, tariff uncertainty and the U.S. dollar. NBER Working Paper No. 34728, National Bureau of Economic Research, Cambridge, MA.

Kim, R. and Vogel, J. (2021). Trade shocks and labor market adjustment. *American Economic Review: Insights*, 3(1), 115–130.

Kline, P., Petkova, N., Williams, H. and Zidar, O. (2019). Who profits from patents? Rent-sharing at innovative firms. *The Quarterly Journal of Economics*, 134(3), 1343–1404.

Levell, P., O'Connell, M. and Smith, K. (2017). The exposure of households' food spending to tariff changes and exchange rate movements. IFS Briefing Note BN213, Institute for Fiscal Studies, London.

Luu, N., Woloszko, N., Causa, O., Arriola, C., van Tongeren, F. and Johansson, Å. (2020). Mapping trade to household budget survey: A conversion framework for assessing the distributional impact of trade policies. OECD Trade Policy Papers No. 244, OECD Publishing, Paris.

McKay, A. and Reis, R. (2016). The role of automatic stabilizers in the U.S. business cycle. *Econometrica*, 84(1), 141–194.

Michelena, G., Ernst, C. and Bertin, P. (2025). Tariffs and labor markets: The employment impact of the recent trade conflict. A multiregional input–output analysis. International Labour Organization / arXiv:2512.11578, December 2025.

Moretti, E. (2010). Local multipliers. *American Economic Review: Papers & Proceedings*, 100(2), 373–377.

Navicke, J., Rastrigina, O. and Sutherland, H. (2013). Nowcasting indicators of poverty risk in the European Union: A microsimulation approach. EUROMOD Working Paper No. EM11/13, ISER, University of Essex.

NIESR (2024). Implications of higher US tariffs for the UK economy. National Institute of Economic and Social Research, London, December 2024.

Office for Budget Responsibility (2025). *Economic and Fiscal Outlook*, March 2025 (tariff scenario box). OBR, London.

Office for National Statistics (2025). UK trade with the United States: 2024. ONS, Newport, 25 April. https://www.ons.gov.uk/economy/nationalaccounts/balanceofpayments/articles/uktradewiththeunitedstates/2024.

Peichl, A. (2016). Linking microsimulation and CGE models. *International Journal of Microsimulation*, 9(1), 167–174.

Resolution Foundation (2025a). Trump tariff turmoil. Resolution Foundation, London, April 2025.

Resolution Foundation (2025b). Saving penalties: Reforming the capital rules in Universal Credit. Resolution Foundation, London, April 2025.

Roberts, J. and Taylor, K. (2022). New evidence on disability benefit claims in Britain: The role of health and the local labour market. *Economica*, 89(353), 131–160.

Society of Motor Manufacturers and Traders (2025). Written submission to the House of Commons Business and Trade Committee inquiry on UK–US trade (reference UST0041). UK Parliament written evidence, https://committees.parliament.uk/writtenevidence/167352/html/.

Rodríguez-Clare, A., Ulate, M. and Vasquez, J. P. (2025). The 2025 trade war: Dynamic impacts across U.S. states and the global economy. NBER Working Paper No. 33792, National Bureau of Economic Research, Cambridge, MA.

Rud, J. P., Simmons, M., Toews, G. and Aragón, F. (2022). Job displacement costs of phasing out coal. IZA Discussion Paper No. 15581.

Sutherland, H. and Figari, F. (2013). EUROMOD: The European Union tax–benefit microsimulation model. *International Journal of Microsimulation*, 6(1), 4–26.

Tax Policy Center (2025). Description of the Tax Policy Center's tariff models. Urban–Brookings Tax Policy Center, Washington, DC.

Traiberman, S. (2019). Occupations and import competition: Evidence from Denmark. *American Economic Review*, 109(12), 4260–4301.

Waugh, M. E. (2019). The consumption response to trade shocks: Evidence from the US–China trade war. NBER Working Paper No. 26353, National Bureau of Economic Research, Cambridge, MA.

World Trade Organization (2025). *Global Trade Outlook and Statistics: Update October 2025.* WTO, Geneva.